

Quantifying the Unknown

A numerical analysis of surprise-based observability of acceptance
bias in reject inference

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Mateo Ordóñez Serna

Berlin, 22.09.2026

Abstract

The Reject Inference (RI) problem concerns the challenge of learning from outcomes that are only observed for accepted applicants, and of using information from rejected applicants whose outcomes remain unknown. This selective observation induces acceptance bias and motivates methods for its identification and correction. This thesis introduces a dynamic perspective on RI by proposing surprise as a signal of acceptance bias. *Surprise* is defined as the discrepancy between the expected performance of a classifier, as estimated from historical data, and its realized performance over subsequent acceptance rounds. The central hypothesis is that this discrepancy contains information about the underlying acceptance bias and therefore provides a means of observing it indirectly. First, the RI problem is formulated within a measure-theoretic framework, making the underlying assumptions explicit. Second, the hypothesis that surprise provides an observable signal of acceptance bias is formally formulated. Third, a controlled simulation study investigates the extent to which surprise reflects the underlying bias. Finally, possible applications of the proposed signal and directions for future research are discussed.

1 Introduction

The Reject Inference (RI) problem concerns learning from outcomes observed only for accepted applicants while leveraging information from rejected applicants whose outcomes remain unknown. This selective observation induces acceptance bias and motivates methods for its identification and correction. Reject Inference has been widely studied in applications such as credit scoring, underwriting, and medical decision systems, as well as from a more abstract perspective as a missing data problem. Although both empirical and theoretical findings have been mixed, its potential impact in high-stakes decision systems continues to motivate the search for the “holy grail” to tackle down the problem.

Rather than proposing a new correction method, this work revisits the RI problem itself from a different perspective. Acceptance decisions are repeatedly based on historically selected data, thereby perpetuating acceptance bias. These accepted applicants subsequently become part of the historical data used to make future decisions. RI is therefore in such sequential scenarios, which are for example typical of a financial institution, inherently a dynamic problem. Although several authors, such as Kozodoi et al. (2025), generate simulation data using such an acceptance loop, the dynamic structure of the loop itself has received little attention as a potential source of information. The intuition fueling this work is straightforward: acceptance bias likely operates as a self-reinforcing mechanism. As a biased selection rule is repeatedly applied, the resulting bias may compound over time or stabilize around a sub-optimal equilibrium.

Then, the central research question is whether the temporal feedback induced by repeated acceptance decisions produces observable information about acceptance bias in subsequently accepted applicants. With this objective this thesis has been structured as follows. First, a literature review is presented, highlighting critiques and methodological shortcomings in the RI literature motivating the following sections and specially highlighting the potential of using a dynamic perspective. Second, responding to the critique of Ehrhardt et al. (2021) regarding the lack of formalism in many RI studies, the credit scoring setting is used to formulate a precise definition of the RI problem, bridging the real world and Math. To do so, Measure theory provides a framework to distinguish clearly between stochastic components and the institution’s deterministic decision rules. Third, a simulation framework based on the well-known two-class Linear Discriminant Analysis (LDA) setting is introduced along an acceptance loop simulation framework to generate the data used throughout this research considering explicitly the dynamic nature of the problem. Fourth, clear testable hypotheses regarding acceptance bias observability are stated. Also *surprise* is introduced as a heuristic quantity intended to indirectly capture acceptance bias. Fifth, the bias of the logistic regression (LR) model used in the acceptance loop and the cross-validation (CV) based cut-off selection mechanism gets studied and characterized through numerical analyses. Finally, this understanding of the acceptance bias is used to enlighten the analysis of *surprise*. This work manages to characterize the acceptance bias, showing that it can be observable. *Surprise* only manages to give a weak signal on this bias, but helps to open an outlook on the possible ways to keep studying the dynamic nature of the RI problem as formulated here.

2 Related work

RI can be formulated as a selective observation problem and studied within the framework of missing-data theory. Accordingly, it raises questions of identification, sample selection, estimation, and evaluation under selection bias. The review of Ehrhardt et al. (2021) identified many structural problems in existing RI methods. Among these issues, one problem stands out in particular: as in Hand and Henley (1993), the conclusion is that for RI to provide information beyond that available from the accepts, additional identifying assumptions or supplementary information are required beyond the observed accepted outcomes and applicant covariates. Furthermore, in a spirit close to Ehrhardt et al. (2021), I believe that being explicit about the assumptions concerning both the data-generating process (DGP) and the methodology is important for rigorously assessing the feasibility, potential, limitations, and possible application fields of an RI approach, as well as for formulating meaningful experimental settings.

Because the central concern of this thesis is not merely whether a particular RI procedure performs well, but under which conditions information about rejected applicants can be obtained from the available data, the literature is reviewed along four dimensions:

- What information from the rejects does the RI method exploit, and under what assumptions does this information identify the target of interest?
- Are the assumptions concerning the data-generating process, observation mechanism, and methodology explicitly stated and theoretically or heuristically justified?
- How do authors evaluate the validity, effectiveness, robustness, and practical applicability of their proposed approach, and what limitations do they identify?
- To what extent does the methodology account for the temporal and feedback structure induced by repeated acceptance decisions, including potential self-reinforcement of selection bias?

All of this information is summarized in Table 1. For each paper, I document the specific **RI Approach** and the **Key Identifying Assumptions** required for the method to perform meaningfully in practice. I aim to minimize implicit assumptions and to formulate them according to the strongest plausible interpretation consistent with the paper. I track whether these assumptions are openly acknowledged and considered (**Assump. Explicit?**), and whether the authors provide a rigorous **Theoretical or Heuristic Justification** for the viability of their approach. These flags as Boolean values are sometimes insufficient, as the authors may consider almost all assumptions while leaving one important assumption implicit. The criterion for setting both to true is whether the most important assumptions were made explicit, either by citing a known methodology from which their assumptions can be deduced or by stating them specifically, *and* whether the narrative of the paper (discussion, experimental design, etc.) is consistent with the stated assumptions and justification.

Finally, I classify the Result Type (theoretical, empirical, or simulated), the data basis (**test pool**) on which the results were shown, if data were used, summarize the paper’s conclusion regarding whether the proposed RI method works, identify the **Main Challenge** limiting the approach, such as computational scalability or error propagation, and indicate whether the framework incorporates a **Dynamic Perspective** to model the RI problem or considers it in their approach.

Table 1: Literature Review

Reference	RI Approach	Key identifying assumptions	Assump. explicit?	Theo./heur. justified?	Result type	Test Pool	Does RI work?	Main challenge	Dynamic persp.
Boyes et al. (1989)	Heckman type PD/AR estimation with augmentation where weights are proportional to population frequencies for behavioral scoring model	MNAR, Bivariate Probit assumptions, FI are rational and may seek to optimize for high balances	✓	✓	Emp., sim.	∅	Unclear, only comparison of decision differences presented	Data required for proposed “behavioral” credit scoring model is hard to obtain	
Hand and Henley (1993)	Estimation of $p(x)$ for further usage in EM based methods	Distributional assumptions, sample of reject region or experts judgement	✓	✓	Theo.	∅	Only with labeled rejects’ sample	Economic costs of sample	
Feelders (2000)	EM algorithm, for rejects explicit expectation over the two classes weighted by the posterior	MAR, parametric classifier	✓	✓	Sim.	A + R	Yes, for small samples and feature space	Parameters scale quadratically with features dimensionality, doubtful whether the DGP can be so simple and unrealistic acceptance rule	
Banasik et al. (2003)	Bivariate Probit for PD and AR equations (BVP)	Errors in modelled equations are correlated, knowledge of relevant variables, MNAR	✓	✓	Emp., sim.	A + R	Moderately if MNAR and conservative FI	Identification of AR and PD equations, correlation between their errors being > 0	
Crook and Banasik (2004)	Augmentation and Extrapolation	For Augmentation like Banasik and Crook (2010), for extrapolation $\mathbb{P}(Y = g X)$ similar in accepts and rejects	✓	✓	Emp.	A + R	No	Like Banasik et al. (2003) and Banasik and Crook (2010)	
Banasik and Crook (2007)	BVP, Augmentation and BVP with Augmentation	For BVP like in Banasik et al. (2003). For Augmentation like Banasik and Crook (2010)	✓	✓	Emp.	A + R	No	Also alike to Banasik et al. (2003) and Banasik and Crook (2010)	
Chen and Astebro (2010)	Bayesian Bound and Collapse method on multinomial over combinations of scoring, repayment and missingness indicator, to then extrapolate to obtain complete sample	MNAR, single cut-off point for whole sample, probability of reject estimable		✓	Emp. with simulated cutoff	AR_s	Yes	Data needed scales strongly with range of scores, unclear how the data is MNAR	
Banasik and Crook (2010)	Augmentation via reweighting using AR model	Accept reject model estimable, enough rare accepts to correct	✓	✓	Emp.	A	No, worse than only accepts	Augmentation inadequate for RI	
Maldonado and Paredes (2010)	SVM with penalization term for low confident predictions (low density region) on unlabeled data as learner for self-training	Linear classification boundaries, low density assumption in unlabeled data, accurate classifier per iteration in self-training	✓	✓	Sim. And Emp.	Sim: A + R Emp: AR_s	Yes	Linearity constraint, high dimensional feature space, handling large amounts of data	
Bücker et al. (2013)	Estimation of unconditional means of features and probability of reject for generative model	Label missingness be independent across observations, MNAR	✓	✓	Theo., sim.	A + R	Yes	Independence condition ² illogical in dynamic context, numerical problems	

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²The condition comes from the cited paper Bücker and Krämer, 2011 in the proof

Table 1: Literature Review

Reference	RI Approach	Key identifying assumptions	Assump. explicit?	Theo./heur. justified?	Result type	Test Pool	Does RI work?	Main challenge	Dynamic persp.
Li et al. (2017)	Semisupervised SVM (S3VM): Rejects guide the decision boundary through low density regions in the features space	Low density regions of features space aligned with known labels separation, MAR			Emp.	$\mathbf{A} + \mathbf{R}_e$	Yes	High dimensional feature space, handling large amounts of data	
Tian et al. (2018)	A Kernel-free Fuzzy Quadratic Surface SVM (FQSSVM) approach consisting of: initial FQSSVM-separation to compute eucl. distance metrics to boundary; outlier detection using local neighborhood affinity to eliminate default outliers with rejects seen as bads from here on; a fuzzy weighting scheme based on class centers, local hesitation, and proximity to the boundary, where potentially good rejects receive small missclassification penalization weights; and convex FQSSVM optimization on both accepted and weighted, temporarily labeled rejected samples	MAR DGP, all rejects as bad appropriate for initial separation. Accepted default borrowers with excellent attributes are "outliers" (accidental defaults) whose elimination improves classification accuracy ³ . Rejects positioned near the initial separating boundary represent potential good borrowers, meaning the separation boundary is assumed to be already near optimal. Quadratic surface a good approximator of "any" separation boundary ⁴			Emp.	$\mathbf{A} + \mathbf{R}_e$	Yes	Unclear dependency on heuristic distance measures, unclear whether results come from the inferred labels, explainability	
Xia et al. (2018)	Contrastive Pessimistic Likelihood (CPLE) with LigthGBM, where the CPLE objective is optimized with fully extrapolated rejects, which are weighted by their PD	From CPLE framework: well-defined likelihood, calibrated PD-estimator over reject region, same features available for rejects and accepts, MNAR. From LightGBM usage: pruning breaks self-reinforcing error loop and PD estimation is almost continuous		✓	Emp.	$\mathbf{A}$	Moderately	Computational costs scale with reject rate, ensuring non-overfitted well calibrated classifier	
Anderson (2019)	Extrapolation to rejects through bayesian network, then build scorecard on extrapolated sample	Knowledge and existence of causal non-cyclic relationships between features. Implicitly MAR	✓		Emp.	$\mathbf{A}$	Moderately	Reliability of causality, dimensionality	

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³This assumption implies a series of assumptions alike to the model presented in 3.2⁴Which cannot be, their argument on taylor approximation is taken to be too general

Table 1: Literature Review

Reference	RI Approach	Key identifying assumptions	Assump. explicit?	Theo./heur. justified?	Result type	Test Pool	Does RI work?	Main challenge	Dynamic persp.
Xia (2019)	Estimate Isolation Forest (IFO) on rejects to label them: outliers as goods, inliers as bad where this classification happens through heuristic setting of Contamination Rate (CR). Then LightGBM on rejects and accepts where missing values of rejects' features are set to 0 and all features are 0-1 normalized	MNAR, acceptance decision process is very good: almost only bads are rejected, as IFO-uncommon rejects are expected to be good payers; bads among the rejects are homogenous. 0 imputation is good regardless the type of feature			Emp.	A	Moderately	Accurately estimating the true latent contamination rate (CR) of the rejects, as the optimal CR varies between datasets and heavily impacts model performance	
Shen et al. (2020)	Most similar Rejects selected through 3-way-decision (3WD) theory based on Mahalanobis distance (MD) to accepts are used jointly with accepts to learn higher level features representations of through Self-Thought-Learning (STL). This representation is used for further supervised learning on accepts	STL will produce features better aligned for credit scoring using <i>any downstream algorithm</i> as it does not assume equal distributions of accepts and rejects. MD serves as proxy to prevent negative transferin STL by including only rejects features alike to the accepts, which allows to induce they assume the data is not MNAR. MD is suitable for a mix of contiguous, binary and categorical data		✓	Emp.	A	Yes ⁵	Extremely High computational costs due to 3WD, explainability and credibility	
Man-cisidor et al. (2020)	Use a generative model with latent features L^6 in form $Y \rightarrow L \rightarrow X$, where rejects are used to train the encoder of X onto L , while being tied by the accepts to be consistent in class conditional prior assumptions $Y \rightarrow L$	Causal markov chain $Y \rightarrow Z \rightarrow X$, distributional assumptions on certain marginals of the system. Implicitly MAR ⁷	✓		Emp.	A	Yes	Numerical stability, sensitivity to initial weights, explainability for regulatory	

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⁵They did however not check if the improvement comes from higher feature representation learning or from RI

⁶They use z as letter in the paper, but here that letter stands for the acceptance status and is thus changed

⁷Effectively they model $P(Y|X, Z = a) = P(Y|X, Z = r)$ assumption

Table 1: Literature Review

Reference	RI Approach	Key identifying assumptions	Assump. explicit?	Theo./heur. justified?	Result type	Test Pool	Does RI work?	Main challenge	Dynamic persp.
Kang et al. (2021)	Semisupervised label propagation: use fully connected graph over all observations, where observations alike form clusters that are assigned a label according to known labels in the cluster, regularized to avoid overfitting through penalization of deviation of initial supervised only label setting and controlling for accepts class imbalance through Borderline-SMOTE. Clusters are calculated through mahalanobis distance	Rejects closest to accepts share labels, larger manifolds with common labels concentrations also share labels. Upsampled points on the borderline between minority and majority contain more valuable information	✓	✓	Emp. Including A/B testing	A, A under new policy	Yes	Computational costs, Borderline SMOTE scales bad with features dimension, explainability for distance metric with discrete features	
Annas et al. (2022)	Semisupervised hidden markov model, where a sequential model along the (binned) features is modeled with two hidden states representing bad and good outcomes. The hidden states are semantically anchored at the beginning with a maximum likelihood estimation of transition parameters in the markov chain. The Rejects are used to finetune the transition probabilities in an unsupervised manner	A Markov Chain along the features in a specific order is sensible and can be meaningfully tied to hidden states, the rejects further do not break up the semantics of the hidden states.			Emp.	A, A + R _e	Yes, on A	Rejects and Accepts must have the same available features, unclear how to define the order of features in chain	
Kang et al. (2022)	Conditional Wasserstein Generative Adversarial Network with Gradient Penalty (CWGAN-GP) combined with Multi-Task Learning (MTL). First, CWGAN-GP models the complete population feature space using rejects with “virtual labels” ⁸ to generate synthetic defaults. Second, a multi-task network with shared embedding layers jointly optimizes default risk on the CWGAN-GP-rebalanced dataset and acceptance classification using the rejected features.	Likely MNAR ⁹ . Same features among accepts and rejects. Virtual label mechanism for rejects does not bias the generative process and does not need weighting these observations. The same latent feature representations for rejects works for modeling GB and AR equations. That specific AR modeling alleviates the bias on GB.		✓	Emp.	A	Yes for rebalancing and moderately prediction ¹⁰	Computational burden of training generative model, selecting the rate of generated samples, explainability	

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⁸The derivation or assignment mechanism for these virtual labels is not explicitly specified in the paper.⁹As they model explicitly the Accept/Reject decision, not explicitly formulated¹⁰The greatest gain in prediction comes from the rebalancing, the MLT part was not benchmarked to other methods considering auxiliary tasks like Heckman’s model

Table 1: Literature Review

Reference	RI Approach	Key identifying assumptions	Assump. explicit?	Theo./heur. justified?	Result type	Test Pool	Does RI work?	Main challenge	Dynamic persp.
Shen et al. (2024)	For Joint Distribution Adaptation (JDA) to reduce dist. Differences between rejects and accepts, and for cost sensitive SVM	JDA can turn MNAR into MAR in latent space, low density separability assumption	✓	✓	Emp.	A	Yes	JDA depends on high feature dimensionality, computational costs, explainability	
Ribeiro and Raimundo (2025)	Confident Inlier Extrapolation (CI-EX): A semi-supervised framework that iteratively identifies non-outlier rejected samples per class using class-conditional Isolation Forest and assigns labels to at most the $c \in \mathbb{N}$ most confident candidates based on probabilities from a balanced ¹¹ classifier	IFO inliers with high confidence is enough to avoid inferring too many wrong labels. Fully unbiasing is not necessarily sensible. IFO is adequate for outlier detection. Implicitly: likely MAR (or close to) ¹²	✓	✓	Emp.	AR_s	Moderately, Trade-off dependent between AUC and Kick-out ¹³	Managing the explicit trade-off between maximizing population debiasing (AUK) and maintaining accepted-set accuracy (AUC), alongside the computational overhead of multi-step iterative retraining, handling same features in rejects and accepts	
Kozodoi et al. (2025)	Bayesian Evaluation: Use random conditional bernoulli based extrapolation on evaluation set iteratively to get bayesian metric. For inference: iteratively partially extrapolate to generate high confidence rejects' labels to generate larger complete data for training scorecard	MAR or MNAR data, partial debiasing is more useful than extrapolating all rejects, rejects contain higher bad rates, posterior estimate can be calibrated ¹⁴	✓	✓	Sim. and Emp.	A + R	Yes for eval., estimation also under specific conditions	BASL: sensitivity to class imbalance, understanding when it is worthwhile using it	

Note: The used abbreviations mean: assump. = assumptions, heur. = heuristically, theo. = theoretical/theoretically, persp. = perspective, emp.=empirical, sim.=simulation, PD = Probability of Default, EM = Expectation Maximization, AR=Accept/Rejects, GB = Goods/Bads. For the Evaluation Pool column: All refer to holdout samples with **A** accepts, **R** rejects, **R_e** extrapolated rejects and **AR_s** are accepts that were divided into 'simulated' accepts and rejects according to some criterion.

Consistent with the findings of Ehrhardt et al. (2021), I find that assumptions are often not made explicit, which I consider harmful for the credibility of the proposals and, in some cases, potentially contradictory. For example, Xia et al. (2018) does not appear to acknowledge that the proposed weighting scheme requires the estimated probability of default (PD) to be (approximately) meaningfully calibrated. They use standard LightGBM, which is not in itself guaranteed to produce calibrated probabilities. Furthermore, it is interesting to find cases such as Chen and Astebro (2010), where missing not at random is assumed explicitly, while the simulated accepts and rejects are generated from the labeled accepts using a missing at random mechanism. This creates a mismatch between the assumed observation mechanism and the mechanism used to construct the simulation.

¹¹In the sense that observations are weighted to account for class imbalance

¹²Otherwise using a scoring model that is calculated without extra assumptions and without estimating AR would be senseless

¹³A RI-directed measure

¹⁴See that this is specially important for the Bayesian Evaluation method, which uses the posterior to repeatedly calculate a metric until convergence

Another case worth highlighting is Annas et al. (2022). They implicitly assume that a Markov chain over an ordered feature sequence can be meaningfully linked to hidden states related to loan outcomes. This relationship is established algorithmically, and the empirical results appear to support the claim. However, they do not discuss the role of the feature ordering or whether this ordering has a substantive interpretation that justifies the Markov assumption. This raises the question of whether the observed success is partly an artifact of the particular feature ordering or experimental design. These observations strengthen my belief in the utility of being explicit about which assumptions are made when studying the RI problem.

Also, Anderson (2019) exemplifies that being explicit about the assumptions does not necessarily mean that they are sufficiently well justified to support an analysis of the proposed approach. See that they do not justify their approach as noted in the second column. In their case, it is assumed that causal relationships between features are acyclic, but it is not argued under which circumstances this should be expected. For example, marital status and asset holdings may be mutually related over time: financial circumstances may influence marriage decisions, while marriage may subsequently influence asset accumulation. In general, I consistently found that a lack of theoretical or heuristic justification weakened the credibility of an approach, which may hinder confidence in RI methods and their adoption in practice.

Nonetheless, being explicit about both assumptions and justification is not foolproof. For example, Shen et al. (2020) motivate the usage of Self-Thought-Learning (STL) for higher-level feature representation by arguing that the accepts and rejects do not need to belong to the same region. However, they also use three-way decision theory to select rejects for STL that are similar to the accepts in the Mahalanobis-distance sense. In other words, they select rejects that could be expected to come from the same feature distribution as the accepts. This partially reintroduces the distributional similarity assumption that the preceding argument seeks to relax.

It also stands out that although many papers introduce the RI problem as an open problem without really promising solutions, a substantial number of papers in Table 1 conclude that their proposed method performs well, at least under the evaluation setting considered. This conclusion is, however, often difficult to assess. On the one hand, some papers such as Li et al. (2017) and Tian et al. (2018) evaluate the RI method using a (partially) extrapolated sample containing also rejects, which can create a circular evaluation procedure. On the other hand, other papers generate artificial rejects from the accepts by using some rule to hide the labels of a subset of the accepts (Chen & Astebro, 2010; Maldonado & Paredes, 2010; Ribeiro & Raimundo, 2025). This amounts to selecting a subset from a population that has already been selected as relatively creditworthy, making the resulting artificial rejects potentially unrepresentative of actual rejects.

Simulation-based work also sometimes uses an unrealistic DGP. For example, Bückner et al. (2013) assumes that applicants are independent and identically distributed (i.i.d.) with respect to their observable characteristics, outcomes, and observation mechanism, which is unlikely to represent a dynamic lending environment. Similarly, Feelders (2000) uses a very abstract decision rule. More generally, unrealistic DGPs are sometimes accompanied by a lack of explicit modeling of how the financial institution’s (FI) reality is translated into a mathematical description of the DGP.

The remaining papers largely evaluate their methods through changes in some classification metric on holdout accepts. This approach is likely too narrow, as it is difficult to generalize such results to an understanding of how an RI-based acceptance policy would perform. One option is to use metrics designed specifically for the RI problem, which appear more adequate for evaluating RI approaches (Kozodoi et al., 2025; Ribeiro & Raimundo, 2025). Another, although difficult to implement, is randomized A/B testing as in Kang et al. (2021), which provides a much more direct empirical test of the consequences of deploying an RI-based policy.

It is also useful to examine the methods that, in my assessment, combine sensible experimental settings, plausible assumptions, and convincing evaluation methods. I identify three in particular: Bayesian Evaluation (Kozodoi et al., 2025), cost-sensitive semi-supervised support vector machine (S3VM) with joint distribution adaptation (JDA) (Shen et al., 2024), and the seminal semi-supervised SVM for credit scoring (Maldonado & Paredes, 2010).¹⁵ Besides Bayesian Evaluation, the other methods rely on additional assumptions, reinforcing the findings of Ehrhardt et al. (2021) and Hand and Henley (1993) on the necessity of further identifying assumptions. In the literature reviewed here, I found no paper that explicitly incorporates the inherent dynamic nature of the RI problem in credit scoring. I believe this aspect may provide a basis for formulating a novel plausible assumption when developing RI methods.

In conclusion, the literature review identified three recurring issues that, in my assessment, weaken a substantial part of the existing research. First is the lack of explicit consideration of which assumptions are needed for an approach to work and the underlying question of why the proposed methodology should be expected to work. To address this, I carefully explain my assumptions on the DGP and make transparent how I have implemented them, including a narrative on why this simulation is a valid way to study the RI problem by tying it to an FI’s point of view (Sec. Section 3 and Section 4).

Second, the literature review suggests that meaningful RI identification generally requires additional assumptions beyond those typically used in credit scoring or standard classification problems. As an answer, I propose the hypothesis of self-reinforcing acceptance bias and investigate whether *surprise* can serve as a diagnostic signal for this bias (Sec. Section 5.1) along with numerical evidence (Sec. Section 5.2).

Third, many papers have proposed solutions to the RI problem since at least Feelders (2000), yet the literature continues to treat RI as a problem with substantial unresolved issues. I speculate that this persistence is partly related to the limited credibility of some evaluation methods, conceptual weaknesses, and limited reproducibility of results. Although a full answer to this shortcoming is not provided here, I give an outlook on how the hypotheses investigated in this thesis may inform the structure and evaluation of RI approaches and thereby help address the issues identified above.

¹⁵See that this model is not the most common benchmark in the literature, which tends to prefer the kernel-based SVM of Li et al. (2017). I still prefer this work as it remains a basis on which later literature develops new methods and because it fulfills the aforementioned conditions.

3 Theoretical background

Having identified the importance of transparent assumptions and following the spirit of Ehrhardt et al. (2021), I begin by formally presenting the Reject Inference (RI) Problem in Credit Risk as understood and approached in this thesis, so that the hypothesis formulation and the theoretical properties of the proposed solution can be precisely formulated. First and also following the approach of Ehrhardt et al. (2021) a detailed description in natural language of the biasing mechanism in FIs is done, so that each concept used can be defined properly avoiding confusions. Second a measure theoretic characterization of the biasing process through its assumed underlying random variables (RV) is given. Third a precise formulation of the missing-data mechanism guided by the definitions of Little and Rubin (2002) is presented, so that the bias which RI attempts to alleviate can be clearly recognized including its source. Finally, a notation of the available data for the FI is done as observed realizations of the described processes. Although the main focus of this thesis is not bettering the formalization approaches, I find it important to be thorough so that future work can properly criticize this thesis.

3.1 Concepts definition

When a financial institution (FI) grants a credit, it has first received information related to an applicant. This information consists of features, some of which are (almost always) used in a scoring model (“model features”), while others—e.g. whether the applicant dresses well in a face-to-face application—are not (“hidden features”). Based on the available information, the FI decides whether to approve the loan. Upon approval by the FI, a financing contract can be concluded, leading to disbursement. A credit agreement after drawdown is termed “accepted”.¹⁶ If an application does not lead to a payout by the FI, it is called “rejected”.¹⁷

If an accepted loan is not repaid on time or in full (as defined by the FI), it is classified as a default (“bad” loan); otherwise, it is classified as “good”. Whether an applicant turns out to be a payer or a non-payer is driven by their individual circumstances¹⁸ as well as purely unpredictable events¹⁹, which I term idiosyncratic shocks. The individual circumstances can be understood as a set of characteristics consisting of the model features, hidden features and additional latent factors, which cannot be observed by the FI. Ignoring idiosyncratic shocks for a moment, the repayment outcome can be thought of as being determined *completely* by this full set of features. In this sense, being a payer or non-payer can be viewed as an underlying property of the applicant, even though it is not directly observable at the time of the decision. Therefore, an applicant is called a *good applicant* if in the absence of an idiosyncratic shock, it would have resulted in a good loan (if the loan was accepted). A *bad applicant* is defined analogously.²⁰

This interpretation does not imply that the FI could construct a perfect prediction model.

¹⁶Since the contract requires the consent (“acceptance”) of both parties.

¹⁷No distinction is made between rejection by the FI and withdrawal by the applicant.

¹⁸e.g. income or health

¹⁹such as a bankruptcy due to a corruption scandal in the applicant’s firm, or the applicant winning the lottery

²⁰Although the usage of bad and good for the loan and for the applicants in a slight different manner might feel confusing, it makes the mathematical notation *much* easier.

Even if all features—including hidden ones—were observable, the functional relationship governing repayment would remain unknown. In practice, both limited observability and unknown dependence prevent perfect prediction.

The bias that reject inference addresses is not yet evident. To see it, consider the acceptance process. At the beginning of its operations, the FI has no data about its own clients. It must therefore rely either on a scorecard trained on data from other markets or on rules based on expert judgment. After sufficient data has been collected, the FI can train a scorecard using its own historical data of *accepted applicants*, since repayment outcomes are only observed for financed loans. As a result, the model is estimated on a subset of the applicant population—presumably one with higher creditworthiness—and then applied to the full population of applicants. The distributional differences between these populations, which have been well documented (e.g. see Kozodoi et al., 2025), are (assumed) to be a central source of bias highlighted in the reject inference literature.

Finally, note that the acceptance process is sequential. Newly accepted applicants generate additional observations, which are incorporated into the historical data and used to re-estimate the scorecard. This, in turn, changes future acceptance decisions because of novel data driven changes in the estimated scorecard. In other words, the decision rule evolves over time, and this time dimension must be considered when modeling the reject inference problem. To capture this, I group time into *acceptance rounds*, where each round begins with a re-estimation of the scorecard. At some acceptance round this cyclic process can be visualized like in Figure 1. There it is specially clear where RI typically plays a role (red arrow), influencing the learning algorithm that returns the scorecard, which impacts the FI's acceptance decision.

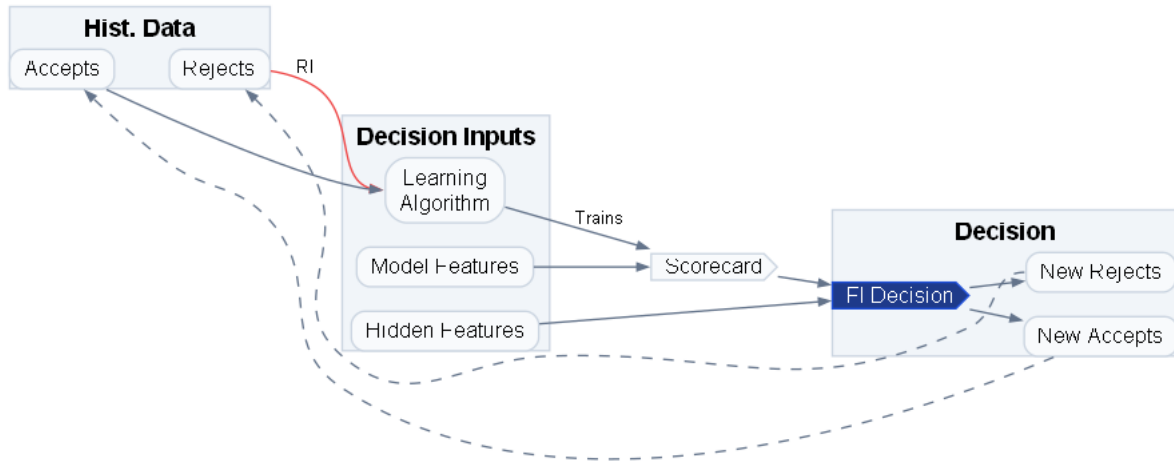


Figure 1: Overview of an FI's acceptance process

3.2 Measure theoretic description

Now with clear concepts from an economic perspective, we can define a stochastic model for the RI problem. I try to be rigorous with the formulation, since I believe it gives a language to precisely identify the hidden assumptions most authors do when investigating RI. However, in some cases I am rather loose on notation and formality -specially in the

notion as a stochastic process-, since I believe some nuances are more confusing than helpful, particularly for a master's thesis.

Let $\mathbf{T} = \{0, 1, \dots, T\} \subseteq \mathbb{N}_0$ be the set indexing the acceptance rounds and $t \in \mathbf{T}$, where $t = 0$ denotes the first time an FI will grant any loans. Further let $X_m, X_h, X_l, Y, Y_m^{(t)}, Z^{(t)}, \epsilon$ and ε be RVs on the probability space $(\Omega, \mathcal{F}, \mathbb{P})$. Define with $j \in \{m, h, l\}$

$$X_j : (\Omega, \mathcal{F}) \rightarrow (\mathbb{R}^{f_j}, \mathcal{B}(\mathbb{R}^{f_j})), \quad f_j \in \mathbb{N}_0$$

as the RVs generating the applicant's model features, hidden features and latent factors²¹. Further see that the case $f_j = 0$ ²² represents a model where X_j always returns an empty tuple, meaning that there are no features of type j . I assume that at least $f_m > 0$. The joint observable feature representation is given by

$$X : (\Omega, \mathcal{F}) \rightarrow (\mathbb{R}^f, \mathcal{B}(\mathbb{R}^f)), \quad f := f_m + f_h \quad (1)$$

with

$$\forall \omega \in \Omega : X(\omega) := \begin{pmatrix} X_m(\omega) \\ X_h(\omega) \end{pmatrix} \quad (2)$$

Further with V a placeholder for $Y, Z^{(t)}$ and $Y_m^{(t)}$ let

$$V : (\Omega, \mathcal{F}) \rightarrow (E_V, 2^{E_V})$$

with 2^{E_V} the power set of E_V and

$$E_Y = \{g, b\}, \quad E_{Z^{(t)}} = \{a, r\}, \quad E_{Y_m^{(t)}} = \{g, b, \mathbf{na}\}$$

be the RVs describing the repayment type, acceptance decision and observable repayment outcome of the loan applied for. Let further $\forall \omega \in \Omega$:

$$Y(\omega) = \begin{cases} b & \text{if applicant is a defaulter} \\ g & \text{if applicant is a payer} \end{cases},$$

$$Z^{(t)}(\omega) = \begin{cases} a & \text{if applicant gets accepted (financed)} \\ r & \text{if applicant gets rejected (non-financed)} \end{cases}$$

and

$$Y_m^{(t)}(\omega) = \begin{cases} Y(\omega) & Z^{(t)}(\omega) = a \\ \mathbf{na} & Z^{(t)}(\omega) = r \end{cases} \quad (3)$$

$Z^{(t)}$ is more precisely modeled as a measurable function of both features sets. Let

$$s_t : \mathbb{R}^{f_m} \rightarrow \mathbb{R}, \quad \mathcal{B}(\mathbb{R}^{f_m}) - \mathcal{B}(\mathbb{R}) \text{ measurable}$$

be a (risk) score function and

$$d_t : \mathbb{R} \times \mathbb{R}^{f_h} \rightarrow \{a, r\}, \quad (\mathcal{B}(\mathbb{R}) \otimes \mathcal{B}(\mathbb{R}^{f_h})) - 2^{\{a, r\}} \text{ measurable}$$

²¹Note that even though not all features are per se continuous and therefore naturally real numbers, a reasonable assumption is that they are representable as real numbers

²²Note: $\mathbb{R}^0 = \{()\}$ is a singleton set containing the empty tuple and $\mathcal{B}(\mathbb{R}^0) = \{\emptyset, \{()\}$

an acceptance decision function. Then

$$\forall \omega \in \Omega : Z^{(t)}(\omega) = d_t(s_t(X_m(\omega)), X_h(\omega)) \quad (4)$$

In this spirit, Y can also be modeled as

$$\forall \omega \in \Omega : Y(\omega) = \begin{cases} \varepsilon(\omega) & \epsilon(\omega) = \text{idiosyncratic shock} \\ \iota(X_m(\omega), X_h(\omega), X_l(\omega)) & \epsilon(\omega) = \text{no idiosyncratic shock} \end{cases} \quad (5)$$

with ι a $\bigotimes_{j \in \{m, h, l\}} \mathcal{B}(\mathbb{R}^{\mathbb{J}})$ - 2^{E_Y} measurable function and

$$\epsilon : (\Omega, \mathcal{F}) \rightarrow (E_\epsilon, 2^{E_\epsilon}), \quad \varepsilon : (\Omega, \mathcal{F}) \rightarrow (E_Y, 2^{E_Y})$$

the RVs defining whether an idiosyncratic shock occurs and its effect (e.g. if there is a shock in the labor market and whether it is positive or negative for the applicant's finances). From this perspective is easy to precisely formulate the difference between a good or bad applicant and a loan. A loan is a bad loan if $Y = b$, an applicant is a bad applicant if $Y = b | \{\epsilon = \text{no idiosyncratic shock}\}$

See that from Eq. 5 in connection with Eq. 4 the FI is modeled to be a “good selector” of features, since all the variables which it uses to make the acceptance decision are assumed to actually play a role in determining Y . Although it might sound like an unrealistic assumption, modeling variables unrelated to Y that affect $Z^{(t)}$ would render reject inference less useful or in some sense easier, as the missingness mechanism would render “closer to ignorable”. An exact formulation of this idea is given in Section 3.3 after Eq. 10.

Now let the joint distribution of the applicants be given by

$$(X_m, X_h, X_l, Y) \sim \mathbb{P}_{X_m, X_h, X_l, Y} \quad (6)$$

defined over the product space of the state spaces of the random variables, with marginals as above.²³ Furthermore I model the applicants' population at each acceptance round $t \in \mathbf{T}$ as an i.i.d. family

$$\left\{ (X_{m,i}^{(t)}, X_{h,i}^{(t)}, X_{l,i}^{(t)}, Y_i^{(t)}) \right\}_{i=1}^{N_t}$$

where each tuple of random variables in the family is distributed according to $\mathbb{P}_{X_m, X_h, X_l, Y}$. Meaning that the features and repayment outcomes of applicants follow a time-invariant distribution.

Therefore, the only source of temporal dynamics relies on the acceptance decision. More specifically it arises from the estimation of s_t . To do so we also need to define the joint distribution of the observable variables to the FI at each acceptance round as

$$(X_m, X_h, Y_m^{(t)}, Z^{(t)}) \sim \mathbb{P}_{X_m, X_h, Y_m^{(t)}, Z^{(t)}}$$

²³Since all feature spaces are Borel spaces, their finite product is also a Borel space. As $(E_Y, 2^{E_Y})$ is finite, the resulting product space satisfies the regularity assumptions ensuring the existence of regular conditional probability kernels (Pollard, 2001, Ch. 4, Theorem 9), which will later be required for conditional modeling arguments.

which also has marginals defined as above.²⁴ From this defining the observable part of the applicant's distribution is straightforward:

$$\left\{ \left(X_{m,i}^{(t)}, X_{h,i}^{(t)}, Y_{m,i}^{(t)}, Z_i^{(t)} \right) \right\}_{i=1}^{N_t},$$

which, conditional on s_t , forms an i.i.d. family for each fixed acceptance round, since d is known. This conditional i.i.d. characteristic already hints that across acceptance rounds the distribution of each family may (and likely does) change. These changes are not exogenous, but dependent on the families of earlier acceptance rounds through estimation of s_t . Further consider that the probability kernels governing the feature variables remain time invariant; the distributional shifts affect only $Z^{(t)}$ and therefore $Y_m^{(t)}$.

This estimation of s_t can and should be further formalized to be transparent on the further assumptions. Let

$$O_{:t} := \bigtimes_{\tau=0}^{t-1} \bigtimes_{i=1}^{N_\tau} \left(\mathbb{R}^{f_m} \times E_{Y_m^{(\tau)}} \right)$$

with $t \in \mathbf{T} \setminus \{0\}$ denote the product of the underlying sets of all past model features and the observable repayment outcomes. $\mathcal{O}_{:t}$ denotes then the corresponding product sigma algebra. For $t = 0$, s_t is assumed to come from some external source (experts opinion, scoring trained on other data). For $t > 0$, s_t is assumed to be generated through a measurable learning procedure

$$\ell_t : (O_{:t}, \mathcal{O}_{:t}) \rightarrow (S_t, \mathcal{S}_t) \quad (7)$$

where the codomain denotes the measurable space of all admissible score functions for time t . See that ℓ_t does not restrict s_t to always be estimated by some specific procedure, such that the score function at one acceptance round could be based on a random forest and in another by a support vector machine.²⁵ Finally, d_t could be formalized in an analogous way to s_t . The only difference is that here the role of the measurable learning procedure would be some measurable acceptance decision where the domain also considers the past score functions. An explicit formalization will not be done, since it does not give any important insights further than the historical dependence already precised by Eq. 7.

3.3 Formalization of the missing data mechanism

Here the data missingness mechanism which gets recorded by $Y_m^{(t)}$ gets formalized according to the canonical definitions of Little and Rubin (2002). Let in this section $t \in \mathbf{T} \setminus \{0\}$ be arbitrary but fixed,

$$X_m^{:t} = \bigcup_{\tau=0}^{t-1} \left\{ X_{m,i}^{(\tau)} \right\}_{i=1}^{N_\tau}$$

and $X_h^{:t}$, $X^{:t}$, $Y^{:t}$, $Y_m^{:t}$ be defined analogously. They represent the “flattened” past random variables in the processes.

²⁴Of course, the marginals of the X_m and X_h components correspond to the marginals according to $\mathbb{P}_{X_m, X_h, X_t, Y}$. Argumentation for regularity is analogous to the former.

²⁵Although an SVM does not deliver a scoring in the traditional sense, there are methods to make it deliver one, for example Platt scaling (Platt et al., 1999) or see the predicitons values -1, 1 as scores themselves.

The three canonical cases of missingness are Missing Completely at Random (MCAR), Missing At Random (MAR) and Missing Not At Random (MNAR). The first case means that the missing data is independent of covariates and *and* labels, meaning

$$\mathbb{P}(Z^{(t)} = r | Y^{:t}, X_m^{:t}, X_h^{:t}) = \mathbb{P}(Z^{(t)} = r)$$

so $Z^{(t)}$ is independent of the three sets. This would be observed if the FI rejected fully at random (e.g. depending on the mood of the employees), a “through the door” acceptance policy (always accept) or would always reject. In any of these three cases, the scoring function s_t would never be biased, meaning its parameters θ are expected to be calculated “correctly”.

Here is important to explain what exactly biasing in this context means, as talking about an “unbiased estimator” for methods like XGB is not the same as for Maximum Likelihood estimators. Biasing is defined here relatively loose: Given knowledge of the statistical properties of the estimation method for s_t and enough data, one has good reasons to believe that the estimated s_t will be a good approximation of some scoring proportional to $\mathbb{P}(Y = b | X)$. In this sense an unbiased data-set occurs when it is representative enough (again, assuming enough data, the small sample problem is not addressed here) to get such a s_t .

MAR on the other hand happens when the missingness only depends on the covariates but not on the outcome variable. Here I define MAR as the case where the missingness depends only on the model features, i.e.

$$\mathbb{P}(Z^{(t)} = r | Y^{:t}, X_m^{:t}, X_h^{:t}) = \mathbb{P}(Z^{(t)} = r | X_m^{:t}) \quad (8)$$

Note that even though this definition is conceptionally the same as in many other sources, it should not be confused with

$$\mathbb{P}(Z^{(t)} = r | Y^{:t}, \underbrace{X_m^{:t}, X_h^{:t}}_{X^{:t}}) = \mathbb{P}(Z^{(t)} = r | X_m^{:t}, X_h^{:t}), \quad (9)$$

which could also be a valid definition of MAR but with a different nuance.

Eq. 9 is a valid definition of MAR in the sense that a sample composed of $Z^{:t}, Y^{:t}, X_m^{:t}, X_h^{:t}$ would be MAR, and specially since for $f_h = 0$ it is equivalent to Eq. 8. However, for $f_h > 0$ using a MAR definition like Eq. 9 would imply that a scoring function s_t does not get trained on MAR data, as $Z^{(t)} | X_m^{:t} \not\perp Y^{:t}$. Considering that RI aims to reduce or eliminate the sample bias to get an unbiased score function, it makes sense to focus on the bias that does concern s_t . Further, within the present framework, Eq. 9 follows directly from the definition of $Z^{(t)}$ -see Eq. 4- as a measurable function of X_m and X_h .

If not only Eq. 8 is fulfilled, but we also have no omitted variable bias, i.e. $f_l = 0$ (there are no latent factors also determining Y) as well, Maximum Likelihood estimators are also consistent under this kind of missingness mechanism (Little & Rubin, 2002), even if the model family is wrong, as Quasi-Maximum-Likelihood consistency results would also apply, rendering ignorable²⁶ the missing data mechanism. Of course if an estimator is consistent, then it also generates an unbiased scoring function in the sense of this thesis.

²⁶Ignorable in the sense of Rubin (1976), i.e. one can train the model using only complete cases and still obtain consistent or unbiased results.

This argument is often used by FIs and critics of RI to defend the usage of classical statistical estimators like LR without needing to consider the data missingness, either assuming $f_l = 0$ or that the model features contain enough information of the latent factors, so that the omitted variable bias is negligible. If the FI collects enough data of the applicant, this might be a somewhat credible claim, but is it really enough to just ignore the missingness mechanism? Later the hypothesis checking section (Section 5.2) will give some perspective on this regard.

The last case (MNAR) is the hardest case to handle, as the missing data cannot be ignored and still be able to train an unbiased scoring function. Formally it can be written as a negation of Eq. 8, i.e.

$$\mathbb{P}(Z^{(t)} = r | Y^{:t}, X_m^{:t}, X_h^{:t}) \neq \mathbb{P}(Z^{(t)} = r | X_m^{:t}) \quad (10)$$

which can be intuitively understood as the acceptance decision being (also) driven by factors not considered by s_t but upon which $Y^{:t}$ does depend on. This is characterized with $f_h > 0$. Note that the dependence of $Y^{:t}$ to $X_h^{:t}$ is necessary for MNAR-ness. If one would use some factor independent of $Y^{:t}$ for the decision (e.g. does the applicant have blond hair?), decisions overwritten by this kind of factors would effectively render a MCAR sample, since in these cases $Z_i^{(t)} \perp\!\!\!\perp \{Y^{:t}, X_m^{:t}\}$.²⁷ Additionally see that this discussion makes it natural to alternatively (and equivalently) write the MNAR condition as

$$Z^{(t)} | X_m^{:t} \not\perp\!\!\!\perp Y^{:t} \quad (11)$$

Furthermore consider that MNAR (at least as modeled here) inherently implies an omitted variable bias for $f_h > 0$, as $Y^{:t}$ depends on $X_h^{:t}$ as well. This makes the evaluation of the effectiveness of RI as a tool to correct for MNAR complicated, as one cannot cleanly separate the effect of MNAR-ness and of the omitted variable on the performance loss of the score-card estimation. At most one can reduce the effect of the omitted variable bias by increasing the dependence structure between X_m and X_h , however this would on the other hand *also* weaken the dependence described by Eq. 11 and in this sense approach MNAR-ness to the MAR case.

²⁷Note: See that this definition of X_h as RVs on which Y depends upon is related to the case discussion of Banasik and Crook (2007) on the variables sets to be used for the acceptance-reject model and the default-repayment model. Such a discussion is however not necessary here given the probabilistic model assumed in this thesis.

4 Simulation framework

Having defined the theoretical background, it is now time to outline a simulation framework that fulfills the conditions established above and allows me to study the hypotheses to be presented in Section 5.1. I use an acceptance loop that is further developed from the loop described in Kozodoi et al. (2025). This section describes the implementation and justifies the main design decisions.

4.1 Applicants' population

Recalling Section 3.2, the applicant population is determined by the joint distribution of the repayment outcome and the features. In the simulation, I do not consider latent features (case $f_l = 0$). I first define a baseline repayment outcome $\tilde{Y} := Y | \{\epsilon = \text{no external effect}\}$ representing the outcome in the absence of an idiosyncratic shock. This variable simplifies the description of the applicant population and provides the cleanest connection to the distinction between good and bad applicants as discussed in the last section.

The chosen DGP of $(\tilde{Y}, X)$ is a two-class Gaussian model that specializes to the classical linear discriminant analysis (LDA) setting through the common-covariance assumption imposed below. Although this is a deliberately simple DGP, its well-known statistical properties make it possible to study the RI problem in a controlled setting.

It holds

$$\tilde{Y} \sim \text{Bernoulli}(\pi_b) \quad (12)$$

with $\pi_b \in (0, 1)$ denoting the probability of a bad applicant. The observable feature vector X^{28} is defined as a Gaussian mixture with two components, a good one (payers) and a bad one (defaulters). With $o \in \{g, b\}$, the class-conditional distributions are characterized as

$$X | \{\tilde{Y} = o\} \sim \mathcal{N} \left(\mu_o = \begin{pmatrix} \mu_{m,o} \\ \mu_{h,o} \end{pmatrix}, \Sigma_o = \begin{pmatrix} \Sigma_{m,o} & \Sigma_{mh,o} \\ \Sigma_{mh,o}^\top & \Sigma_{h,o} \end{pmatrix} \right)$$

with Σ_o , $\Sigma_{m,o}$, and $\Sigma_{h,o}$ strictly positive definite (and therefore non-singular). Letting X_o denote $X | \{\tilde{Y} = o\}$, I also use ϕ_{X_o} to denote its probability density function. Since X_o is normally distributed, the density is

$$\phi_{X_o}(x) := \frac{1}{\sqrt{(2\pi)^f |\Sigma_o|}} \exp \left\{ -\frac{1}{2} (x - \mu_o)^\top \Sigma_o^{-1} (x - \mu_o) \right\}.$$

For further analysis, it is useful to write down the joint distribution of $(X, \tilde{Y})$, which corresponds to the marginal distribution of $\mathbb{P}_{X_m, X_h, X_l, Y}$ over the latent feature component,

²⁸Recall from Eq. 2: $X = (X_m^\top, X_h^\top)^\top$.

conditional on the absence of an idiosyncratic shock. Its density²⁹ is defined as the Radon-Nikodym derivative

$$p_{X,\tilde{Y}} = \frac{d\mathbb{P}_{X,\tilde{Y}}}{d(\lambda^f \otimes \#_Y)}$$

where $\#_Y$ denotes the counting measure over E_Y . Evaluated pointwise,

$$p_{X,\tilde{Y}}(x, o) = \pi_o \cdot \phi_{X_o}(x) \quad (13)$$

where $\pi_g := 1 - \pi_b$.

From this, the marginals follow directly. The marginal mass function of $\tilde{Y}$ is

$$p_{\tilde{Y}}(o) = \int_{\mathbb{R}^f} p_{X,\tilde{Y}}(x, o) \lambda^f(dx) = \pi_o, \quad (14)$$

which corresponds to the probability mass function of a Bernoulli distributed variable. The marginal density of X is

$$p_X(x) = \int_{E_Y} p_{X,\tilde{Y}}(x, y) \#_Y(dy) = \pi_b \phi_{X_b}(x) + \pi_g \phi_{X_g}(x) \quad (15)$$

and is therefore the density of a Gaussian mixture. Hence,

$$\mathbb{P}(\tilde{Y} = o \mid X = x) = p_{\tilde{Y}|X}(o \mid x) = \frac{\phi_{X_o}(x) p_{\tilde{Y}}(o)}{p_X(x)}. \quad (16)$$

When the class-conditional covariance matrices are equal, the posterior in Eq. 16 has the familiar logistic form. Consequently, LR is correctly specified for $\mathbb{P}(\tilde{Y} = b \mid X = x)$ in this setting (Hastie et al., 2009, Ch. 4). Under standard regularity conditions, the maximum-likelihood estimator is then consistent. Since LR is also widely used in practical credit scoring applications, this provides an additional reason to choose the present DGP.

I next introduce idiosyncratic shocks into the model. As defined from Eq. 5 onwards, ϵ indicate whether an idiosyncratic shock occurs and ε the outcome generated under a shock. Recall that these variables are independent of $(X, \tilde{Y})$ and from one another. Both of these RVs are assumed to be Bernoulli distributed with

$$\mathbb{P}(\epsilon = \text{no shock}) = \pi_\epsilon > 0 \quad \text{and} \quad \mathbb{P}(\varepsilon = b) = \pi_{\varepsilon,b} \in (0, 1).$$

The resulting posterior probability of a bad outcome is

$$\mathbb{P}(Y = b \mid X = x) = p_{Y=b|X}(x) = \pi_\epsilon p_{\tilde{Y}|X}(b \mid x) + (1 - \pi_\epsilon) \pi_{\varepsilon,b} \quad (17)$$

²⁹Following Pollard (2001, Ch. 3, Sec. 1), densities are understood in the Radon-Nikodym sense with respect to a dominating measure.

which defines the Bayes-optimal conditional probability of a bad outcome.

Since the FI does not use X_h when constructing the scorecard, the score function s_t can only approximate $\mathbb{P}(Y = b \mid X_m = x_m)$. This distribution can be obtained by integrating out X_h . Writing $x = (x_m^\top, x_h^\top)^\top$, Eq. 13 gives

$$p_{X_m, \tilde{Y}}(x_m, o) := \int_{\mathbb{R}^{f_h}} p_{X, \tilde{Y}}(x, o) \lambda^{f_h}(dx_h).$$

Consequently,

$$X_{m,o} := X_m \mid \tilde{Y} = o \sim \mathcal{N}(\mu_{m,o}, \Sigma_{m,o}) \quad (18)$$

and therefore

$$\mathbb{P}(Y = b \mid X_m = x_m) = p_{Y=b \mid X_m}(x_m) = \pi_\epsilon \underbrace{\frac{\phi_{X_m,b}(x_m) p_{\tilde{Y}}(b)}{p_{X_m}(x_m)}}_{=: p_{\tilde{Y} \mid X_m}(b \mid x_m)} + (1 - \pi_\epsilon) \pi_{\epsilon,b} \quad (19)$$

which is the Bayes-optimal conditional probability available to a scorecard restricted to X_m .

For the simulation, I choose a deliberately simple DGP so that its statistical properties are easy to understand. Specifically, $f_m = 2$, $f_h = 1$, and $\Sigma_g = \Sigma_b = \Sigma$, so that the class-conditional distributions differ only through their means. This common-covariance specification is the LDA setting used throughout the simulation. I also set $\pi_\epsilon = 1$, so that no idiosyncratic shocks are considered.³⁰ The common covariance matrix is parameterized as

$$\Sigma(\sigma_{m,12}, \sigma_{mh}) = \Sigma = \begin{pmatrix} 1 & \sigma_{m,12} & \sigma_{mh} \\ \sigma_{m,12} & 1 & \sigma_{mh} \\ \sigma_{mh} & \sigma_{mh} & 1 \end{pmatrix} \quad (20)$$

and the means are given by

$$\mu_b = \begin{bmatrix} 0.0 \\ 0.0 \\ 0.0 \end{bmatrix}, \quad \mu_g = \begin{bmatrix} 1.0 \\ 2.0 \\ -1.0 \end{bmatrix}. \quad (21)$$

The form in Eq. 20 allows to vary the dependence structure within the chosen standardized covariance parameterization while maintaining positive definiteness. The admissible values of $\sigma_{m,12}$ and σ_{mh} can therefore be selected to study the possible DGPs induced by them.

³⁰Although this excludes idiosyncratic shocks from the base simulation, the mechanism is retained in the general DGP because it provides a useful extension. Under symmetric shocks, it would leave the decision boundary unchanged while increasing the conditional variance of $Y \mid X$. That would complicate the classification problem, but it would be irrelevant to the study of RI as a dynamic problem. Specially considering that LR as the scorecard to be used is robust to class outliers (Hastie et al., 2009, Ch. 4)

A useful diagnostic is the performance of the classifiers defined by Eq. 17 and Eq. 19. These are the Bayes classifiers³¹ for $Y | X$ and $Y | X_m$, respectively. Their theoretical performance is measured by the Bayes error rate, defined as the probability that the classifier makes an incorrect prediction when classifying as bad whenever the predicted probability of bad exceeds the predicted probability of good, and as good otherwise.³²

The Bayes error rate can be calculated analytically using Eq. 42, as derived in the appendix (see Section 7.3.2.2). Let P_e^* denote the error rate of the full model following Eq. 17, and let P_e^m denote the error rate of the reduced model following Eq. 19. In general,

$$0 \leq P_e^* \leq P_e^m,$$

and, under the present covariance parameterization, P_e^m does not depend on σ_{mh} because integrating out X_h leaves the marginal distribution of X_m unchanged. I can therefore examine the Bayes error rates over the complete admissible region of the covariance parameters, which can be characterized analytically using Eq. 38 (see Section 7.3.1 in the appendix).

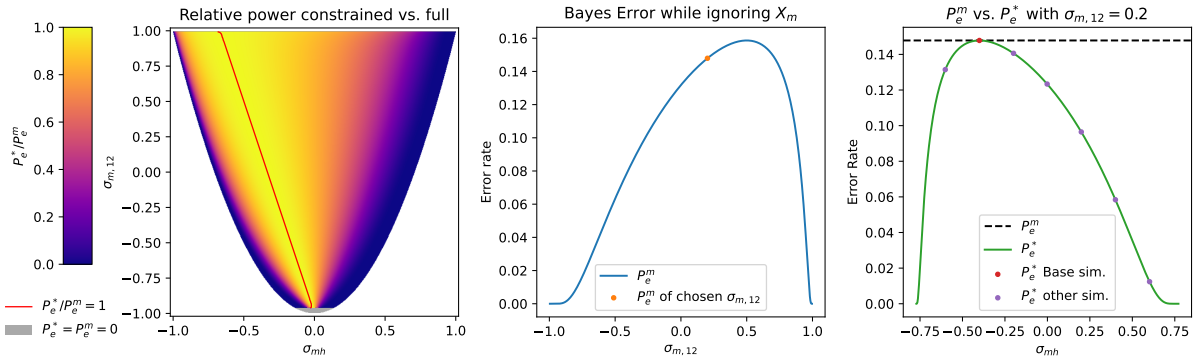


Figure 2: Analysis of Bayes Errors distributions for admissible covariance matrices

In the left panel of Figure 2 we see the heat map of the ratio P_e^*/P_e^m . The graphic shows numerically that

$$\forall \sigma_{m,12} \exists \sigma_{mh} : P_e^* = P_e^m. \quad (22)$$

For many of these cases, Eq. 22 does not imply the Bayes error rate being zero. As to be disclosed in Section 4.2, the MNAR mechanism is introduced through overwrites based on extreme values of X_h . Such a mechanism creates dependence of the acceptance decision on a hidden variable, which would generate an omitted variable bias when estimating a scorecard based on the model variables. The cases in Eq. 22 are useful because they allow to study this MNAR effect while keeping the achievable Bayes classification error of the reduced model equal to that of the full model. Thus, X_h can remain marginally associated with the repayment outcome while providing essentially no additional discriminatory information for the Bayes classification boundary once the model features X_m are given.

³¹Also known as the generative model in the LDA literature.

³²This rule does not include a reject-option or abstention class.

The joint distribution (Y, X_m) is itself an LDA-type DGP.³³ Consequently, an appropriately specified LR can consistently estimate through maximum likelihood its conditional class probability under the usual regularity conditions.

These observations motivate the choice of $\sigma_{m,12}$ for which Eq. 22 holds without the Bayes error rates being zero. To study the effect of the omitted variable, it is useful to choose a value of $\sigma_{m,12}$ for which the ratio P_e^*/P_e^m varies substantially over the admissible values of σ_{mh} . At the same time, P_e^m should remain sufficiently high that the classification problem is not trivial.

As depicted in the middle panel of Figure 2, I set $\sigma_{m,12} = 0.2$. I then choose σ_{mh} using the right panel. For the base simulation, I select a value for which the contribution of X_h to the full Bayes boundary is negligible, namely $\sigma_{mh} = -0.4$. For the base simulation, this gives $P_e^* = P_e^m = 14.78\%$. Other values of σ_{mh} used later in the sensitivity analysis are selected on an equidistant grid from -0.6 to 0.6.

The complete base DGP is visualized in Figure 3. The Perfect Bayes Boundary corresponds to the classification boundary of the classifier defined by Eq. 17. This boundary is derived in the appendix (see Section 7.3.2.1) using the equal covariance. It can be written as the hyperplane $\mathcal{D}^*$

$$w^\top x - \beta = \log c \quad (23)$$

with

$$w = \Sigma^{-1}(\mu_b - \mu_g),$$

$\beta = w^\top(\mu_b + \mu_g)/2$, and $c > 0$ a constant that depends on the idiosyncratic-shock parameters π_ϵ and $\pi_{\epsilon,b}$ as well as the unconditional probability of default π_b .

³³This follows from Eq. 18 together with the fact that integrating out X_h leaves the common class-conditional covariance structure for X_m .

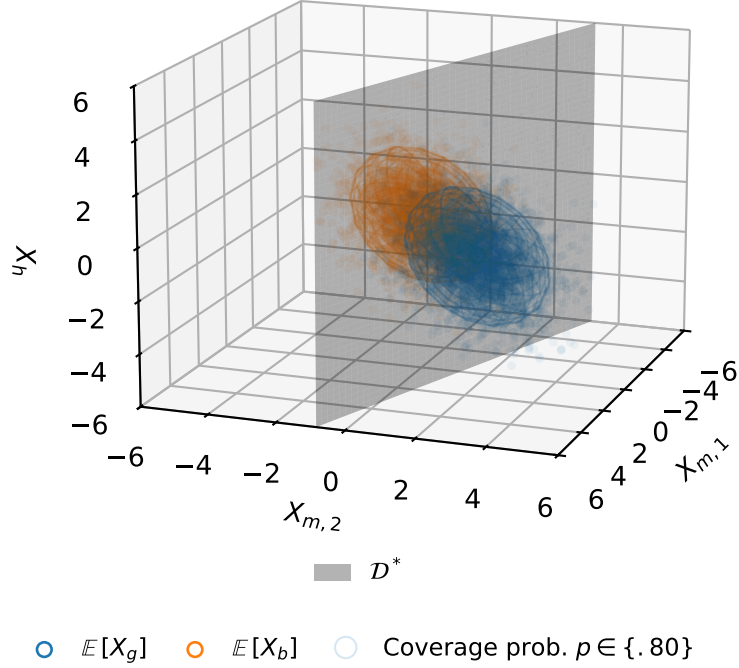


Figure 3: Representation of joint distribution (Y, X)

Considering the Bayes error rate of the problem and the good separation achieved by the hyperplane of the Perfect Bayes Boundary visible in Figure 3, there is further reason to expect LR to estimate the separation boundary accurately when all relevant variables are available. To understand how the problem changes when one variable is omitted, a pairwise component plot such as Figure 4 is useful.

Two decision boundaries are shown there. The first is the projection of $\mathcal{D}^*$ into a two-dimensional plane, obtained by fixing the component not shown in the pairwise plot at its unconditional mean. For a component indexed by r , this gives

$$x[\mathbf{r}] = \mathbb{E}(X|\epsilon = \text{no shock})[\mathbf{r}] = (\pi_b)\mu_b[\mathbf{r}] + (1 - \pi_b)\mu_g[\mathbf{r}].$$

The second boundary, denoted by $\mathcal{D}^r$, is obtained by integrating out the non-shown component from the joint distribution and then deriving the corresponding Bayes decision boundary.

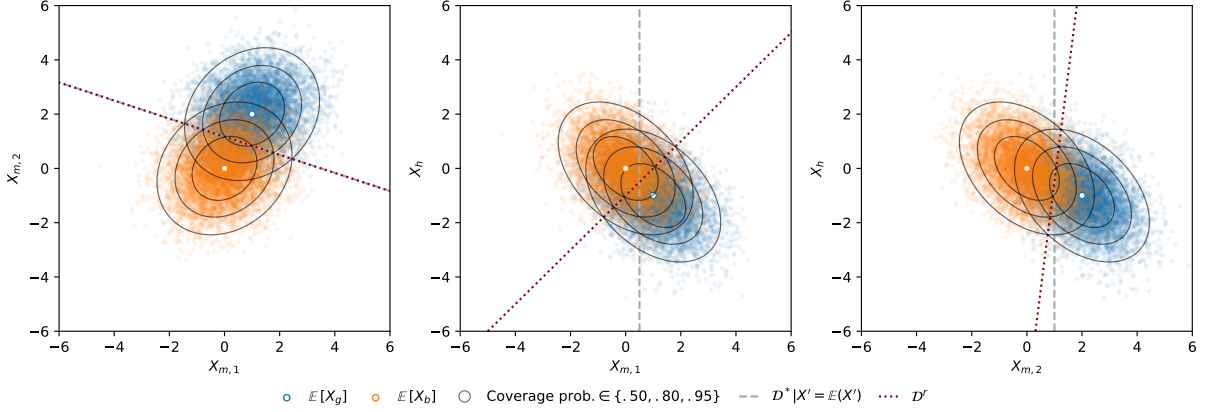


Figure 4: Pairwise covariates component representation of Y, X

Through Figure 4, the limited contribution of X_h to the Perfect Bayes classifier can be visualized. In the left panel, the projected $\mathcal{D}^*$ and $\mathcal{D}^r$ coincide when only X_m is considered. In the middle and right panels, the projections of $\mathcal{D}^*$ are almost parallel to the X_h axis, indicating that the full-information Bayes classifier effectively ignores X_h . At the same time, the corresponding $\mathcal{D}^r$ boundaries differ from the projected full-information boundary. Thus, although the full-information decision boundary assigns essentially no coefficient to X_h , X_h remains marginally informative about the repayment outcome.

These observations depend only weakly on the exact value used for the projection. The reason is that the component of the normal vector corresponding to X_h is numerically zero, namely $-2.22 \cdot 10^{-16}$. The realizations of X_h also almost always lie within the interval $[-6, 6]$ ³⁴, so the contribution of X_h to the classification boundary is negligible. These properties further clarify the usefulness of the DGP for introducing an MNAR mechanism based on X_h without simultaneously inducing substantial omitted-variable bias in the classification boundary.

The pairwise plot also indicates that the classification problem should be well approximated by LR using only the two model features X_m . At the same time, the plots containing X_h illustrate how its relationship with the model features changes the marginal separability. In particular, the boundary shown in the panels containing X_h has a positive slope with respect to $X_{m,1}$ and $X_{m,2}$. A higher positive correlation between X_h and the model features therefore increases the separation available when X_h is observed, while a negative correlation should have the opposite effect.

As a final remark, it is important to make the sampling mechanism of $\mathbb{P}_{X,Y}$ transparent because the rest of the study is primarily numerical. The simulation implementation should therefore be open to scrutiny. The sampling logic is summarized in Algorithm 1.

First, the realizations of Y are sampled from the Bernoulli distribution defined in Eq. 12. These labels are then reordered so that observations from the two classes can be stored in contiguous memory. The corresponding samples from X_b and X_g are then generated and

³⁴This interval comes from a visual inspection of the data, not a proper definition of a range where the probability of X_h lying there would be very big. However given the class conditional means and the unit variance of X_h one could expect such a “properly calculated” interval to lie within $[-6, 6]$

stacked in the same order as the labels. This preserves the label-feature correspondence while allowing an efficient implementation.

Algorithm 1 Sampling from $\mathbb{P}_{X,Y}$

Note: Here the random variables and their realizations are represented with the same notation for convenience

```

1: procedure CREDITDGP.SAMPLE( $\sigma_{m,12}, \sigma_{mh}, \pi_b, n$ )
  1. Sampling good and bad components while ensuring contiguous memory per class
2:    $\forall i \in \{1, \dots, n\} : B_i \stackrel{iid}{\sim} \text{Bernoulli}(\pi_b) \in \{0, 1\}$  with  $\mathbb{P}(B_i = 1) = \pi_b$ 
3:    $n_b \leftarrow \sum_{i=1}^n B_i, n_g \leftarrow n - n_b$ 
4:    $\mathbf{Y} \leftarrow (\underbrace{b, \dots, b}_{n_b \text{ times}}, \underbrace{g, \dots, g}_{n_g \text{ times}})$ 
 $\triangleright \mathbf{Y} \in \{b, g\}^n$ 

  2. Multivariate Gaussian sampling for goods and bads
5:    $\Sigma_{chol} \leftarrow \text{CholeskyDecomposition}(\Sigma(\sigma_{m,12}, \sigma_{mh}))$ 
 $\triangleright$  See Eq. 20

6:   for  $o \in \{b, g\}$  do
7:      $\mathbf{m}_o \leftarrow (\mu_o, \dots, \mu_o)^\top \in \mathbb{R}^{n \times f}$ 
8:      $\mathbf{E} \leftarrow (E_1, \dots, E_n)^\top, \forall j \in \{1, \dots, n\} : E_j \stackrel{iid}{\sim} \mathcal{N}(0_f, I_f)$ 
 $\triangleright$  Done via i.i.d. sampling of  $n \cdot f$  univariate standard normals

9:      $\mathbf{X}_o \leftarrow \mathbf{m}_o + \mathbf{E}\Sigma_{chol}^\top$ 
10:   end for
11:    $\mathbf{X} \leftarrow \begin{pmatrix} \mathbf{X}_b \\ \mathbf{X}_g \end{pmatrix}$ 
12:   return  $\mathbf{X}, \mathbf{Y}$ 
13: end procedure

```

4.2 The acceptance loop

After defining the underlying applicant distribution, the remaining parts of the DGP can be characterized through the implementation of the acceptance loop.³⁵ A detailed representation is given in Algorithm 2.

The key dynamic feature of the simulation is that applicants in each round are generated independently from the same underlying DGP. Temporal dependence is introduced through the acceptance mechanism and the accumulation of accepted observations used to estimate subsequent scorecards. Thus, the applicant population itself is stationary, while the observed training population evolves through repeated selection - as required in the model under Section 3.2.

At initialization, the acceptance policy is based on a business rule that is consistent with the true DGP: higher values of the second component of X_m tend to indicate good applicants because $\mu_g[1] > \mu_b[1]$ Eq. 21. After initialization, acceptance is based on a scorecard and a CV-based decision function.

The LR scorecard is selected because, under the equal-covariance Gaussian specification, its functional form matches the conditional class probability. Consequently, LR can recover the population decision boundary implied by $\mathbb{P}(Y = b \mid X_m)$ under correct specification. Therefore, under MAR and usual regularity conditions LR can be consistently estimated to deliver the necessary parameters to obtain said decision boundary. The LR scorecard

³⁵See Figure 1 for a low-detail overview of the acceptance loop.

Algorithm 2 Acceptance Loop

```

1: procedure ACCEPTANCELOOP( $\sigma_{m,12}, \sigma_{mh}, k, CV, \text{opt\_crit}, p_{\text{bias}}$ )
Require:  $\sigma_{m,12}, \sigma_{mh}, \pi_b$  fulfilling the conditions above,  $p_{\text{bias}} \in [0, 1]$ 
Require:  $\text{opt\_crit}$  a criterion for calculating the acceptance threshold,  $k \geq 2, CV \geq 1$ 
2:    $\pi_b \leftarrow 0.5$ 
3:    $DGP \leftarrow \text{CreditDGP}(\pi_b, \mu_b, \mu_g, \Sigma(\sigma_{m,12}, \sigma_{mh}))$   $\triangleright$  See definitions in Eq. 12, 20 and 21
4:    $n_i \leftarrow 6000, f_m \leftarrow 2, f_h \leftarrow 1, f \leftarrow f_m + f_h, n \leftarrow 1000$ 
5:    $\mathbf{X}, \mathbf{Y} \leftarrow DGP.\text{sample}(n = n_i)$   $\triangleright \mathbf{X} \in \mathbb{R}^{n_i \times f}, \mathbf{Y} \in \{b, g\}^{n_i}$ , Sampling follows Alg. 1
6:    $q_{\text{init}} \leftarrow 0.2$ 
7:    $\tilde{\tau}_{\text{init}} \leftarrow \text{quantile}_{1-q_{\text{init}}}(\mathbf{X}[:, f_m - 1])$   $\triangleright$  Empirical upper quantile
8:    $\mathbf{Z} \leftarrow (a \text{ if } \mathbf{X}[j, f - 1] > \tilde{\tau}_{\text{init}} \text{ else } r \mid j \in \{0, \dots, n_i - 1\})$   $\triangleright \mathbf{Z} \in \{a, r\}^{n_i}$ 
9:    $\mathbf{G} \leftarrow \vec{0}_{n_i}$   $\triangleright \mathbf{G} \in \mathbb{N}_0^{n_i}$ 
10:  for  $t \in \{1, \dots, T\}$  do  $\triangleright T = 300$  in the experiments
11:     $\mathbf{N}_a \leftarrow \{j \in \mathbb{N}_0 \mid j < \text{nrows}(\mathbf{X}), \mathbf{Z}[j] = a\}, n_a \leftarrow |\mathbf{N}_a|$ 
12:     $\mathbf{X}_{\text{hist}}^{(t)} \leftarrow (\mathbf{X}[j, :f_m] \mid j \in \mathbf{N}_a)$   $\triangleright \mathbf{X}_{\text{hist}}^{(t)} \in \mathbb{R}^{n_a \times f_m}$ 
13:     $\mathbf{Y}_{\text{hist}}^{(t)} \leftarrow (\mathbf{Y}[j] \mid j \in \mathbf{N}_a)$   $\triangleright \mathbf{Y}_{\text{hist}}^{(t)} \in \{b, g\}^{n_a}$ 
14:     $s_t \leftarrow \text{trainLR}(\mathbf{X}_{\text{hist}}^{(t)}, \mathbf{Y}_{\text{hist}}^{(t)})$   $\triangleright$  LR trained with Newton-Raphson
15:     $\tau_t \leftarrow \text{CalcCVThreshold}(\mathbf{X}_{\text{hist}}^{(t)}, \mathbf{Y}_{\text{hist}}^{(t)}, k, CV, \text{opt\_crit})$   $\triangleright \tau_t \in [0, 1]$ , see Alg. 3
16:     $\mathbf{X}_{\text{new}}^{(t)}, \mathbf{Y}_{\text{new}}^{(t)} \leftarrow DGP.\text{sample}(n = n)$ 
17:     $\mathbf{Z}^{(t)} \leftarrow (z_0, \dots, z_{n-1})^\top$ , where  $z_j = a$  if  $s_t(\mathbf{X}_{\text{new}}^{(t)}[j, :f_m]) < \tau_t$ , else  $z_j = r$ 
18:    if  $p_{\text{bias}} > 0$  then  $\triangleright$  Apply the MNAR mechanism
19:       $x_b \leftarrow \text{quantile}_{1-\frac{p_{\text{bias}}}{2}}(\mathbf{X}_{\text{new}}^{(t)}[:, f - 1])$ 
20:       $x_g \leftarrow \text{quantile}_{\frac{p_{\text{bias}}}{2}}(\mathbf{X}_{\text{new}}^{(t)}[:, f - 1])$ 
21:       $\forall j \in \{0, \dots, n - 1\} : \mathbf{Z}^{(t)}[j] \leftarrow \begin{cases} a & \mathbf{X}_{\text{new}}^{(t)}[j, f - 1] < x_g, \\ r & \mathbf{X}_{\text{new}}^{(t)}[j, f - 1] > x_b, \\ z_j & \text{otherwise.} \end{cases}$ 
22:    end if
23:     $\mathbf{X} \leftarrow \begin{pmatrix} \mathbf{X} \\ \mathbf{X}_{\text{new}}^{(t)} \end{pmatrix}, \mathbf{Y} \leftarrow \begin{pmatrix} \mathbf{Y} \\ \mathbf{Y}_{\text{new}}^{(t)} \end{pmatrix}, \mathbf{Z} \leftarrow \begin{pmatrix} \mathbf{Z} \\ \mathbf{Z}^{(t)} \end{pmatrix}, \mathbf{G} \leftarrow \begin{pmatrix} \mathbf{G} \\ \vec{t}_n \end{pmatrix}$ 
24:  end for
25: end procedure

```

is therefore estimated at line 14 using the Newton-Raphson algorithm to maximize the logistic likelihood.

The outcome Y is generally allowed to depend on both X_m and X_h . In the base simulation, however, the effect of X_h on the Bayes classification boundary is negligible, as shown above, while idiosyncratic shocks are excluded. Thus, in the absence of MNAR overwrites ($p_{\text{bias}} = 0$), the reduced LR model is correctly specified for the conditional relationship between Y and X_m .

The initialization rule is based on q_{init} , so the initial number of accepted observations is determined by the corresponding empirical upper quantile, not by p_{bias} . Consequently, when the initial sample is sufficiently large (which is probably given here as the initial accepted sample has $n_i \cdot q_{\text{init}} = 1200$ observations), the first LR estimate should be close to the population conditional model.

Algorithm 3 CV-based threshold calculation

```

1: procedure CALCCVTHRESHOLD( $\mathbf{X}, \mathbf{Y}, k, \text{CV}, \text{opt\_crit}$ )
Require:  $\text{opt\_crit} \in \{\text{ks}, \text{roc}\}$ ,  $\mathbf{X} \in \mathbb{R}^{n \times c}$ ,  $\mathbf{Y} \in \{g, b\}^n$ 
2:    $\mathbf{TH} \leftarrow []$ 
                                      $\triangleright$  Buffer for repetition-level thresholds
3:   for  $\text{cv} \in \{1, \dots, \text{CV}\}$  do
4:      $\mathbf{TH}_{\text{cv}} \leftarrow []$ 
                                      $\triangleright$  Buffer for fold-level thresholds
5:      $\{\mathbf{I}_{tr,1}, \dots, \mathbf{I}_{tr,k}\}, \{\mathbf{I}_{va,1}, \dots, \mathbf{I}_{va,k}\} \leftarrow \text{CVIndexSet}(n, k)$ 
                                      $\triangleright$  For all  $k$ :  $\mathbf{I}_{tr,k} \cap \mathbf{I}_{va,k} = \emptyset$  and  $\bigcup_{\kappa} \mathbf{I}_{va,\kappa} = \{0, \dots, n-1\} = \mathbf{I}_{tr,k} \cup \mathbf{I}_{va,k}$ 
6:     for  $k \in \{1, \dots, k\}$  do
7:        $\mathbf{X}_{tr} \leftarrow (\mathbf{X}[i] \mid i \in \mathbf{I}_{tr,k})$ ,    $\mathbf{Y}_{tr} \leftarrow (\mathbf{Y}[i] \mid i \in \mathbf{I}_{tr,k})$ 
8:        $\mathbf{X}_{va} \leftarrow (\mathbf{X}[i] \mid i \in \mathbf{I}_{va,k})$ ,    $\mathbf{Y}_{va} \leftarrow (\mathbf{Y}[i] \mid i \in \mathbf{I}_{va,k})$ 
9:       if Less than 4 observations of each label in the validation or training set then
10:        continue
11:       end if
12:        $l \leftarrow \text{trainLR}(\mathbf{X}_{tr}, \mathbf{Y}_{tr})$ 
13:        $n_{va} \leftarrow \text{nrows}(\mathbf{X}_{va})$ ,    $\mathbf{N}_{va} \leftarrow \{0, \dots, n_{va} - 1\}$ 
14:        $\mathbf{S} \leftarrow \text{OrderAscending}\{(l(\mathbf{X}_{va}[j]) \mid j \in \mathbf{N}_{va})\}$ , with  $\mathbf{Y}_{va}$  reordered accordingly
15:       if  $\text{opt\_crit} = \text{ks}$  then
16:          $\forall o \in \{b, g\}, i \in \mathbf{N}_{va}: F_o[i] \leftarrow \frac{\sum_{j=0}^i I(\mathbf{Y}_{va}[j]=o)}{i+1}$ 
                                      $\triangleright$  Empirical class CDFs
17:          $i^* \leftarrow \arg \max_{i \in \mathbf{N}_{va}} |F_b[i] - F_g[i]|$ 
18:         else
                                      $\triangleright$  ROC-optimal point
19:          $\forall i \in \mathbf{N}_{va}: \text{ROC}[i] \leftarrow (FPR_{\tau=\mathbf{S}[i]}, TPR_{\tau=\mathbf{S}[i]})^\top$ 
                                      $\triangleright$  False Positive Rate and True Positive Rate on  $(\mathbf{X}_{va}, \mathbf{Y}_{va})$  if threshold was  $\mathbf{S}[i]$ 
20:          $i^* \leftarrow \arg \min_{i \in \mathbf{N}_{va}} \|\text{ROC}[i] - (0, 1)^\top\|_2$ 
21:         end if
22:          $\mathbf{TH}_{\text{cv}} \leftarrow \text{append}(\mathbf{TH}_{\text{cv}}, \mathbf{S}[i^*])$ 
23:       end for
24:        $\mathbf{TH} \leftarrow \text{append}(\mathbf{TH}, \text{ArithmeticMean}(\mathbf{TH}_{\text{cv}}))$ 
25:     end for
26:      $\tau^* \leftarrow \text{ArithmeticMean}(\mathbf{TH})$ 
                                      $\triangleright$  Returns na if insufficient labels are available
27:   return  $\tau^*$ 
28: end procedure

```

The decision function therefore has two stages. First, an acceptance threshold is estimated

from the scorecard through repeated k -fold cross-validation and a chosen optimization criterion (line 17). The threshold procedure is summarized in Algorithm 3.

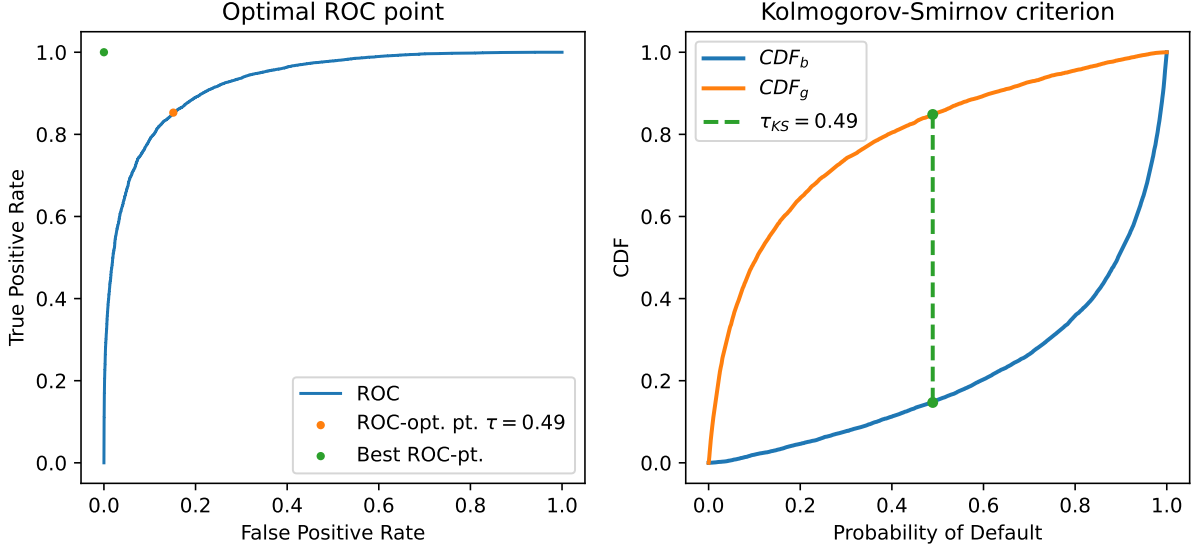


Figure 5: Visualization of criteria for CV-based threshold calculation

The procedure is intended to stabilize threshold estimates. For each of the CV repetitions, the accepted historical sample is divided into k training-validation pairs. An LR model is fitted on each training set, and a threshold is selected on the corresponding validation set. Two threshold criteria are considered. The first is the ROC threshold whose point is closest to $(0, 1)$ in Euclidean distance (ROC-optimal point). The second is the threshold maximizing the Kolmogorov-Smirnov statistic, namely the maximum absolute difference between the empirical class CDFs. The thresholds are then averaged within each repetition and across repetitions. Both criteria are visualized in Figure 5.

In the second stage of the decision process, the MNAR mechanism can optionally be applied. This occurs when $p_{\text{bias}} > 0$. The overwrite rule depends on the tails of the hidden feature $X_h^{(t)}$. More specifically, applicants in the lower tail are accepted and applicants in the upper tail are rejected, regardless of the scorecard decision. Because $\mu_g[2] < \mu_b[2]$, the lower tail of X_h is enriched for good applicants and the upper tail for bad applicants. The overwrite mechanism therefore tends to accept good applicants and reject bad applicants, while still allowing overlap between the two classes.

The parameter p_{bias} controls thereby the total mass assigned to the lower and upper tails of X_h to which the overwrite rule applies. It is therefore a control parameter for the strength of the hidden-feature-based selection mechanism, not necessarily the exact fraction of observations whose original scorecard decisions are changed. So line 21 of Algorithm 2 should not be understood as a mechanism flipping every decision in for the extreme values of the hidden factor.

5 Using the acceptance dynamics

With the simulation framework defined, I can now formulate how the acceptance dynamics may be used to study decision bias. The central hypothesis is that, under suitable conditions, acceptance bias may leave a detectable signal in the observed outcomes of future accepts. I first specify what I mean by **bias observability** and introduce *surprise* as a candidate diagnostic signal. I then describe a setting in which such a signal should not systematically arise. Based on this reasoning, I formulate testable hypotheses and use the simulation to investigate them. Finally, I discuss how the resulting insights could inform the construction of RI approaches.

5.1 Hypothesis definition

To formalize the statement that bias should be observable and potentially reduced, I first need to specify what is understood as bias and how it could be observed. I begin by introducing *surprise* as a candidate diagnostic signal, motivated by the known effects of sample selection on the evaluation of predictive models. I then describe a setting in which such behavior should not systematically arise in the dynamic RI setting studied here (recall Section 3). Finally, based on these considerations, I formulate the hypothesis and the evaluation rationale used to study it.

5.1.1 Surprise signal

Sample selection can distort the evaluation of predictive models when the distribution of the evaluation sample differs from the distribution represented in the data used to estimate model performance (Sugiyama et al., 2007). In credit scoring, Kozodoi et al. (2025) illustrate this issue and propose a framework to correct the resulting evaluation metric. This raises the question of whether a similar discrepancy can be observed by comparing accepted applicants across subsequent acceptance rounds. Concretely, the performance estimated from historical accepts may differ from the realized performance of the same scorecard on accepted applicants in a subsequent round. I use this discrepancy as a candidate signal of the changing selection process.

Let $m_{CV}^{(t)}$ denote the cross-validated estimate of a performance metric m obtained from the historical accepted sample available before round t . I define the surprise $\eta^{(t)}$ at acceptance round t as the difference between this estimate and the realized performance of the scorecard on the newly accepted applicants:

$$\eta^{(t)} := m_{CV}^{(t)}(\mathbf{X}_m^{:t}, \mathbf{Y}^{:t}) - m(s_t; \mathbf{X}_m^{(t)}, \mathbf{Y}^{(t)}).$$

For the metrics used in the numerical analysis, larger values indicate better performance. Hence, a positive value of $\eta^{(t)}$ means that the historical performance estimate is higher than the realized performance on future accepts, so a discrepancy in the same overoptimistic flavor as reported by Kozodoi et al. (2025). For the analysis below, I also discuss its absolute value, which I call the **absolute surprise** and denote by $|\eta^{(t)}|$.

5.1.2 An unsurprising setting

The claim that acceptance bias may be observable from accepted applicants alone is strong and should therefore be motivated. The purpose of this section is not to provide a rigorous proof, but to develop an intuition for a setting in which systematic surprise should not be expected.

Consider an FI operating under the following assumptions:

1. The applicant population distribution is time invariant.
2. The number of applicants N_t is sufficiently large in each round such that each new accepted sample contains enough observations for the performance metric to be estimated with limited sampling variation.
3. The scorecard learning procedure is unchanged across rounds, i.e. the same learning algorithm ℓ is applied in every round.
4. The cutoff-selection procedure is unchanged across rounds and is applied according to the same criterion.
5. The missing-data process satisfies the MAR assumption³⁶ for $t \in \mathbf{T}$ and $i \in \{1, \dots, N_t\}$.

Under MAR, the acceptance decision is conditionally independent of the repayment history given the observable model-feature history. Therefore,

$$Z^{(t)} \perp\!\!\!\perp Y^t \mid X_m^t$$

which implies that, conditional on the observable features, the selection mechanism does not additionally depend on the repayment outcomes. Together with a stationary applicant population and stable learning and cutoff procedures, this means that the distribution of accepted applicants should remain approximately stable across rounds.

In such a setting, the scorecard is repeatedly trained and evaluated on samples generated by essentially the same acceptance mechanism. Consequently, the cross-validated performance estimated from historical accepts should remain informative about performance on future accepts. Systematic large deviations between the two are therefore not expected, so $\eta^{(t)}$ should remain small apart from finite-sample variation.

This observation motivates the interpretation of surprise as a diagnostic signal. Persistent or systematically large values of $\eta^{(t)}$ suggest that at least one of the assumptions above is violated. Assumptions 1 to 4 are enforced by the simulation framework. Consequently, within the simulated environment, persistent surprise cannot be attributed to changes in the applicant population, the learning procedure, or the cutoff-selection procedure. Instead, it is expected to primarily reflect model misspecification and departures from MAR. Surprise is therefore a natural candidate signal for diagnosing such departures.

³⁶Recall Eq. 8.

5.1.3 Hypothesis formulation and evaluation rationale

The surprise signal is generated by the complete acceptance policy at each round t . This policy comprises the learning algorithm ℓ_t , its resulting scorecard s_t , and the decision function d_t . I hypothesize that the accepts-only policy described in the acceptance loop Section 4.2 will generate comparatively large $|\eta^{(t)}|$. In contrast, a policy with access to the complete history of labeled applicants should generate smaller $|\eta^{(t)}|$ values.

The hypothesized mechanism is sequential. At any given round, assuming a reasonably functioning acceptance policy, the historical accepts already represent a region of the applicant population with comparatively high creditworthiness. This is the information available to the FI about the complete applicant population. The acceptance process can therefore tend to select applicants from regions that are estimated to have still higher creditworthiness in subsequent rounds. Repeatedly applying this process can progressively narrow the acceptance region.

Consequently, historical accepts may cover a broader region of the applicant population than accepts in later rounds. This changes the marginal distribution of the accepted covariates across rounds and creates a form of covariate shift between historical and future accepted samples. The performance estimated on the historical sample may therefore differ from the realized performance on future accepts, causing $|\eta^{(t)}|$ to increase.

If these hypotheses are correct, I expect the following behaviors:

1. The progressive overfocusing on a region of the applicant population should manifest itself in decreasing thresholds for the score-based decision stage over subsequent acceptance rounds,³⁷ which should in turn produce decreasing acceptance rates.
2. Under the dynamic MNAR mechanism used in the simulation, the coefficients of the LR scorecard should increasingly deviate from the corresponding population parameters of the LR model based only on X_m . Since MAR is violated, increasing the number of observations does not guarantee convergence towards these parameters. In the present LDA setting, these population parameters correspond to the Bayes decision boundary of the model reduced to the observable model features X_m .
3. The absolute surprise should increase during the early rounds and subsequently either continue increasing or stabilize at a non-negligible level.

To meaningfully compare these behaviors, I also define a full-information benchmark that provides access to the fully labeled history of all applicants. This benchmark differs from the accepts-only policy only in the information available for estimating the scorecard and the threshold. The scorecard remains an LR based only on the model features, the threshold is estimated with the same procedure shown in Algorithm 3, and the acceptance decision is taken in the same way: first through the threshold-based decision and, if $p_{\text{bias}} > 0$, through the same MNAR mechanism.

The full-information benchmark therefore shows how the FI would make its decisions if complete outcome information were available. It is an oracle-like benchmark rather than a

³⁷Recall that the score is defined as a risk score and is estimated to be proportional to the conditional PD. Therefore, lower thresholds correspond to decisions that accept applicants with a lower estimated probability of default and hence higher estimated creditworthiness.

practical RI method and as such should be seen as the best an RI method could perform. In particular, when $p_{\text{bias}} > 0$, the scorecard still uses only X_m and therefore remains subject to omitted-variable effects from the hidden feature X_h .

5.2 Numerical evidence of the bias observability as surprise

5.2.1 Overfocusing mechanism

The hypothesized overfocusing mechanism should be reflected in both the evolution of the acceptance thresholds and the resulting distribution of accepted applicants. I therefore examine both aspects for the two threshold-selection metrics. Threshold evolution can be assessed directly from the right panels of Figure 6. The distributional change is less direct to observe. One proxy is the relationship between the number of accepted applicants in a round and the corresponding bad rate among accepts. Although this relationship does not contain the covariates themselves, the simple DGP and the linear decision boundary under MAR³⁸ make it informative about the region of the applicant population selected by the policy. The pairwise feature distributions in Figure 4 provide the corresponding reference.

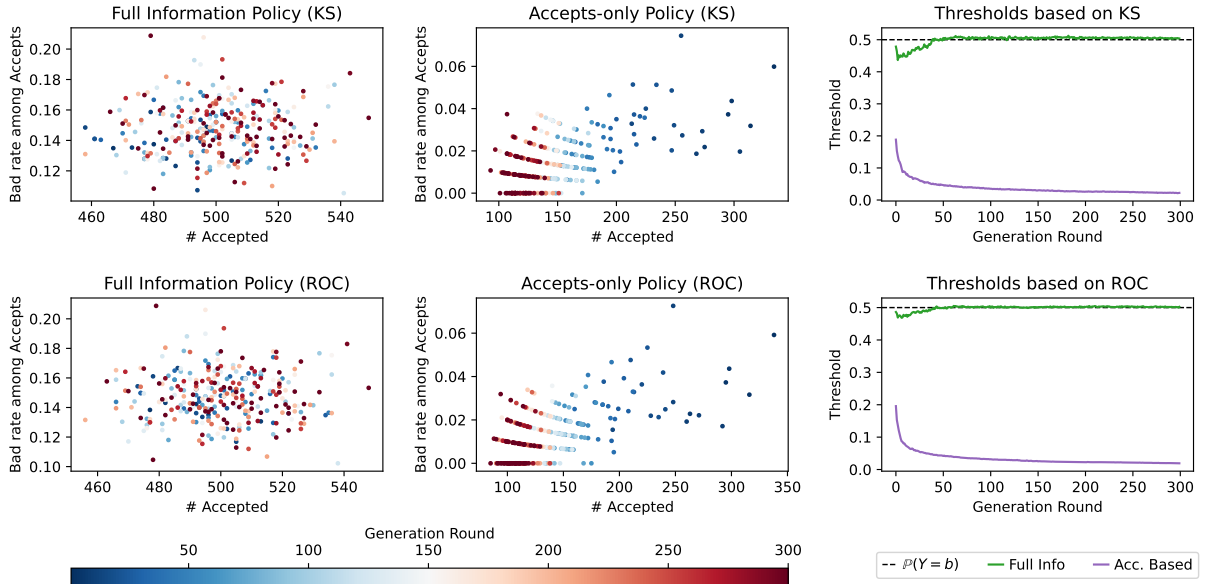


Figure 6: Summary of acceptance behavior across rounds in base simulation

The first high-level observation of Figure 6 is that the metric chosen for threshold estimation has no perceivable effect on the threshold evolution, the acceptance rate, or the relationship between acceptance rate and bad rate among accepts.³⁹ This shows that the qualitative results are robust to the choice between the two threshold-selection metrics, which is unsurprising because both depend on the scorecard’s rank-ordering capability.

The threshold evolution depicted in the right panels of Figure 6 notably shows a rapid decline in the thresholds of the accepts-only policy. The trajectory is approximately

³⁸See that under MAR the selection is fully threshold driven, which given LR estimates with only linear terms, it implies decision hyperplane-boundary.

³⁹At every round, exactly 1000 new applicants are sampled, so the x-axis can be interpreted both in terms of accepted applicants and as an acceptance rate.

hyperbolic: the decrease is initially sharp and then stabilizes after only a few rounds. In contrast, the thresholds of the full-information policy remain very stable and close to $\mathbb{P}(Y = b)$, as expected for a full-information LR in the LDA setting.

The threshold behavior is accompanied by the expected changes in the other panels. For the accepts-only policy, the middle panel shows a rapid decline in the acceptance rate over subsequent rounds. This suggests that the declining thresholds are not primarily caused by a global downward shift in the predicted PDs. Such a shift would represent a calibration change rather than the progressive concentration of the acceptance region. Instead, the increasing concentration of points in the lower-left region of the plane indicates that the policy increasingly selects regions associated with lower predicted PD. This is particularly notable because neither the scorecard estimation nor the threshold-selection procedure explicitly optimizes an economic risk-aware objective or an acceptance volume.

A lower bad rate among accepts can be desirable, but the accompanying decline in the acceptance rate is not necessarily desirable. In the present DGP, where $\mathbb{P}(Y = b) = 0.5 = \mathbb{P}(Y = g)$ and the Bayes error rate of both the restricted and full model is 14.78%, an acceptance rate of 15% or less corresponds to an overly selective policy.

The left panels show that this progressive concentration does not occur for the full-information policy. There is no recognizable temporal pattern in the relationship between acceptance rate and bad rate among accepts. The acceptance rate remains close to 50%, consistent with the stable threshold near $\mathbb{P}(Y = b) = 0.5$, while the bad rate among accepts remains close to the Bayes error rate. Thus, under MAR, the full-information policy behaves as expected from a cost-insensitive Bayes classifier based on X_m .

The next question is how the MNAR overwrites affect this behavior. I first keep $\sigma_{mh} = -0.4$ fixed and vary only the proportion of decisions driven by the MNAR mechanism, p_{bias} . This reduces the effect of the hidden-variable relationship itself and isolates the effect of the overwrite frequency. The simulation therefore considers

$$p_{\text{bias}} \in \{0.05, 0.10, 0.20, 0.35\}. \quad (24)$$

These values produce no substantial change in the threshold trajectory.⁴⁰ The threshold estimates show some variation with p_{bias} and may increase slightly, but their functional form remains essentially unchanged.

By contrast, the evolution of the bad rate among accepts and the acceptance rate shows more noticeable changes. In Figure 7 the effect of changing p_{bias} is shown for the full-information policy (left panels) and the accepts-only policy (right panels), using the thresholds obtained from the optimal-ROC point criterion. The KS criterion produces essentially the same results.

The full-information policy is only weakly affected by increasing p_{bias} . The acceptance rate becomes somewhat more stable around 50%, but the effect is small. The bad rate among accepts changes similarly little and remains close to its baseline value. Overall,

⁴⁰These results are not reported here because the differences are small. Corresponding plots are available in the presentation on slide 24, where Figure 6 for different configurations of σ_{mh} and p_{bias} can be seen. This presentation can be generated from `thesis/presentation/presentation.qmd` in the GitHub repository.

the MNAR overwrites have only a limited effect on the full-information policy in this configuration.

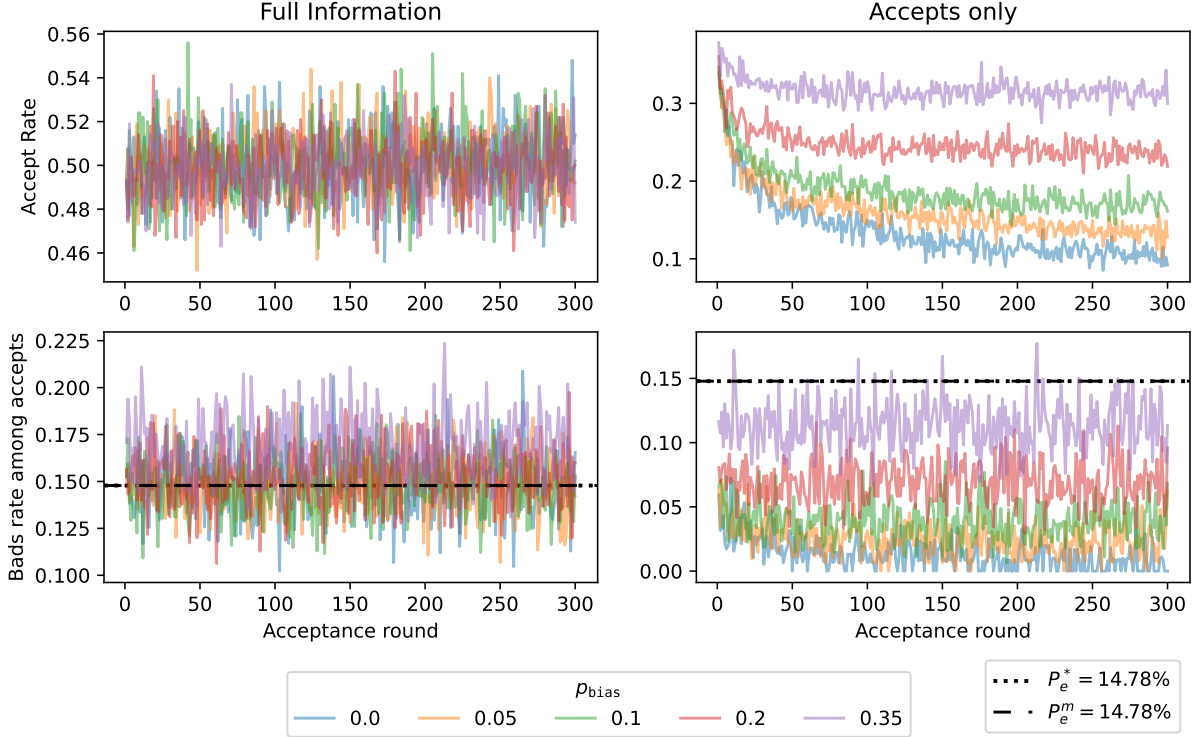


Figure 7: Development of acceptance rates and bad rates among accepts across p_{bias} with $\sigma_{mh} = -0.4$

In contrast, the effect of p_{bias} on the accepts-only policy is substantial. Increasing p_{bias} raises both the number of accepted applicants and the bad rate among accepts, although the increase in the acceptance rate is stronger. The acceptance-rate trajectories change from the pronounced decline observed without overwrites to trajectories that are closer to a constant level. With increasing p_{bias} , the acceptance rate also stabilizes earlier and varies less across neighboring rounds.

The bad rate among accepts shows a similar change. Without overwrites, it follows an approximately hyperbolic decline and stabilizes relatively early. Once the MNAR overwrites are activated, the initial decline becomes less pronounced and is barely visible at higher overwrite proportions. Thus, in this simulation, the effect of the overwrite mechanism on the acceptance rate is stronger than its effect on the bad rate among accepts.

These effects are likely driven by the fact that even for $p_{\text{bias}} = 0.1$ the number of accepted applicants almost doubles, although the estimated thresholds change little. A substantial fraction of acceptance decisions is therefore generated by the overwrite mechanism. Since the same overwrite rule is applied to both policies, an increasing share of their decisions is driven by the same mechanism, making their accepted samples more similar.

When moving away from the base case, where the hidden factor provides little additional information beyond X_m , the qualitative behavior remains similar, especially under MAR.

As X_h becomes more informative about Y , reflected by a larger difference between the restricted and full Bayes error rates $P_e^m - P_e^*$, the rapid decline in the acceptance rate becomes more pronounced, while the bad-rate trajectory changes less visibly. The overfocusing mechanism is therefore also present in the other applicant-population configurations.

In these configurations, the effect of the overwrites becomes more pronounced. For the full-information policy, increasing p_{bias} lowers the bad rate among accepts while the acceptance rate remains close to $P(Y = b)$. For the accepts-only policy, the resulting trajectories move closer to those of the full-information policy. Moreover, the differences between the two policies become more visible as P_e^* decreases. This is not surprising either. With the hidden factor being more informative, the amount of overwrites that do not deliver the same result as the threshold based decision increases, because X_h delivers more clearly distinct information than the obtainable through X_m . Additional examples are provided in the appendix under Section 7.4.

Overall, the numerical analysis provides evidence for the overfocusing mechanism under the specified simulation framework. The effect generally becomes stronger over subsequent acceptance rounds, which is consistent with the self-reinforcing selection mechanism hypothesized above. The MNAR overwrite mechanism considered here, however, reduces the apparent difference between the accepts-only and full-information policies because an increasing fraction of decisions is driven by the common overwrite rule. The particular overwrite specification is therefore less informative for studying realistic manual overwriting behavior: it increasingly bypasses the score-based decision rather than perturbing it through a distinct alternative decision mechanism. This bypassing represents a specially large proportion of the loan granting decisions in the accepts only policy, which may be not realistic in real world scenarios.

5.2.2 LR Parameters convergence

The preceding analysis shows that overfocusing is observable in the accepts-only policy across the simulated configurations, with the effect generally becoming stronger over subsequent rounds. Although the threshold plays an important role, the scorecard itself may also be affected by the dynamics and therefore contribute to the evolution of the acceptance policy. Since the scorecard is always an LR, I examine this effect through its coefficients. I first specify the notation.

A logit model based on model features can be written as

$$p(y = b|x_{1,m}, x_{2,m}) = \frac{1}{1 + \exp \{ -(\beta_0 + \beta_1 x_{1,m} + \beta_2 x_{2,m}) \}}$$

with $p(y = b|x_{1,m}, x_{2,m})$ denoting the conditional probability of default and $(x_{1,m}, x_{2,m})^\top \in \mathbb{R}^{f_m}$ representing a realization of X_m . Under correct model specification, MAR, and standard regularity conditions, maximum likelihood estimators are consistent when the observed data satisfy the required stability conditions. This classical result cannot be transferred mechanically to the present dynamic setting, because the acceptance mechanism changes the distribution of the observations entering the training sample across rounds. Moreover, finite-sample consistency does not imply that the parameter estimates are already close to their population values.

Therefore, to study the behavior of the scorecard in this dynamic environment, I compare the accepts-only estimates with those obtained from the full-information policy. See that the population values can be calculated since the applicant’s distribution is fully known in the experiments. However this would present a comparison that does not consider that the training sample size increases with the acceptance round. To prepare this analysis I first examine the parameters of the full-information policy’s scorecard to assess how much stabilization can be observed at the sample sizes used in the experiments.

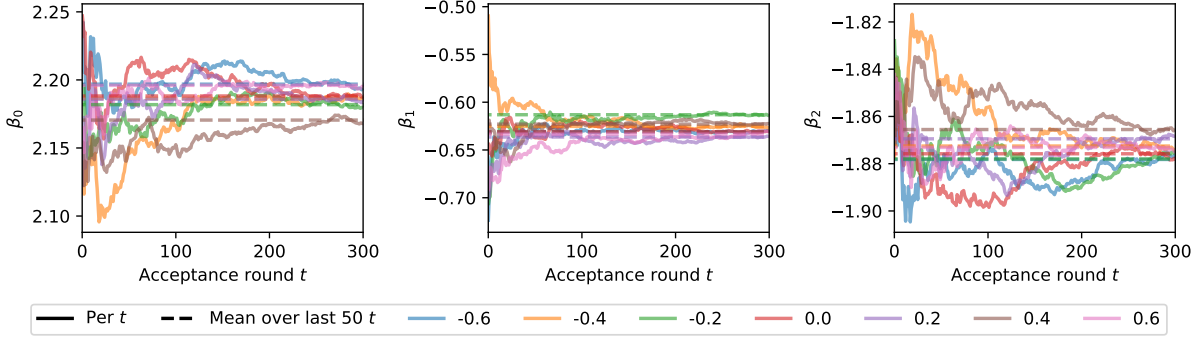


Figure 8: Estimated LR-Parameters on full history for all σ_{mh}

Figure 8 shows the results of this analysis for the simulated applicant populations. The parameter estimates vary only little after the initial rounds, indicating rapid stabilization at the sample sizes considered here. The different applicant-population specifications also converge to similar parameter values, as indicated by the dashed lines. These lines are considered here as approximations of the population parameters towards which the coefficients should be converging to and are calculated as the mean over the values of the last 50 rounds. This is consistent with the fact that the configurations differ only in the covariance between the model features and the hidden feature.

This similarity is useful for the next analysis. To assess whether the accepts-only parameter estimates approach the values observed for the full-information policy, I compare them at each acceptance round. As mentioned before, I use the full-information estimates rather than the population parameter to reduce the effect of finite-sample differences between the simulation configurations. The relative deviation from the full-information parameter is defined as

$$\Delta\beta_k = \frac{\beta_k^{full} - \hat{\beta}_k}{|\beta_k^{full}|}$$

for $k \in \{0, 1, 2\}$, where β_k^{full} is the parameter estimated with full information and $\hat{\beta}_k$ is the corresponding parameter estimated using only accepts.

I plot these deviations in Figure 9, both for the accepts of the accepts-only policy (Biased Acc.), denoted $\hat{\beta}_k^b$, and for the estimates obtained from the accepts of the full-information policy, denoted $\hat{\beta}_k^f$. The latter provides a benchmark for the effect of sampling variation when the accepted sample is generated by the full-information policy. The relative-deviation representation is useful here because the full-information parameter estimates are similar across the simulated population specifications. For reference, I also report the mean of the final-portion full-information parameter estimates as $\bar{\beta}_k^{lim}$.

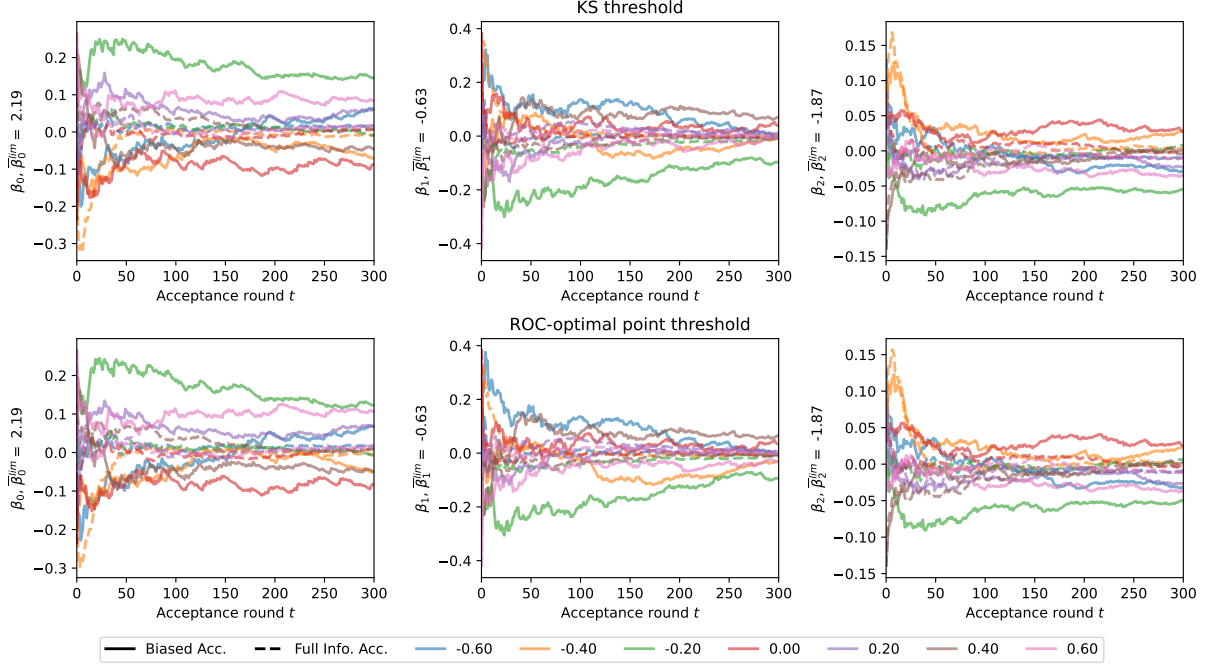


Figure 9: Percentual deviation of Accepts based LR parameters to full information parameters under MAR

From Figure 9 it is immediately clear that the deviations follow essentially the same pattern for both threshold-selection criteria across the simulated applicant populations. This robustness with respect to the threshold criterion also persists across the simulated values of p_{bias} . The deviations of $\hat{\beta}_k^f$ are considerably smaller and more stable than those of $\hat{\beta}_k^b$, and most of them are already close to zero by the 50th acceptance round.

The cleaner convergence-like behavior of $\hat{\beta}_k^f$ cannot be explained solely by the size of the training sample. The accepts-only policy has much lower acceptance rates than the full-information policy: under MAR, the former reaches roughly 10%, whereas the latter remains close to 50%. Even if the accepts-only policy had this low acceptance rate from the beginning, by round 250 its accepted sample would be comparable in size to the full-information sample at round 50. Nevertheless, the deviations of $\hat{\beta}_k^b$ remain substantially larger at $t \geq 250$ than those of $\hat{\beta}_k^f$ at $t = 50$. This indicates that sample size alone is unlikely to explain the difference.

The accepts-only samples also contain substantially fewer bad observations and are therefore more imbalanced. This provides another potential source of instability, but it does not fully explain the persistent differences observed across the configurations. In particular, even in the base configuration, the deviations of $\hat{\beta}_0^b$ and, to a lesser extent, $\hat{\beta}_2^b$ remain substantial after many rounds. For $\hat{\beta}_0^b$, the estimates do not appear to approach the full-information counterpart over the simulated horizon.

These results should nevertheless not be interpreted as evidence that the LR MLE is inconsistent in the dynamic problem. Consistency is an asymptotic property, and the simulations necessarily remain finite. The more defensible conclusion is that the usual consistency argument does not by itself imply that acceptance bias becomes negligible at

realistic sample sizes. By contrast, the much smaller deviations of $\hat{\beta}_k^f$ suggest that obtaining labels for a more representative accepted region could improve scorecard estimation. This provides motivation for investigating RI methods that can recover at least part of this information.

The same methodology can be used to study the effect of p_{bias} on the parameter deviations. The optimal-ROC point criterion is used for all p_{bias} configurations in Figure 10. In almost all configurations, excluding the base case discussed below, increasing p_{bias} increases the relative deviations of the accepts-only and full-information parameters. For most parameters and configurations, the estimates then appear to stabilize at a non-zero distance from the full-information parameters. The deviations of $\hat{\beta}_k^f$ remain consistently smaller than those of $\hat{\beta}_k^b$, suggesting that a more representative accepted sample can reduce parameter distortion even when the MNAR mechanism itself is not explicitly modelled by the scorecard model.

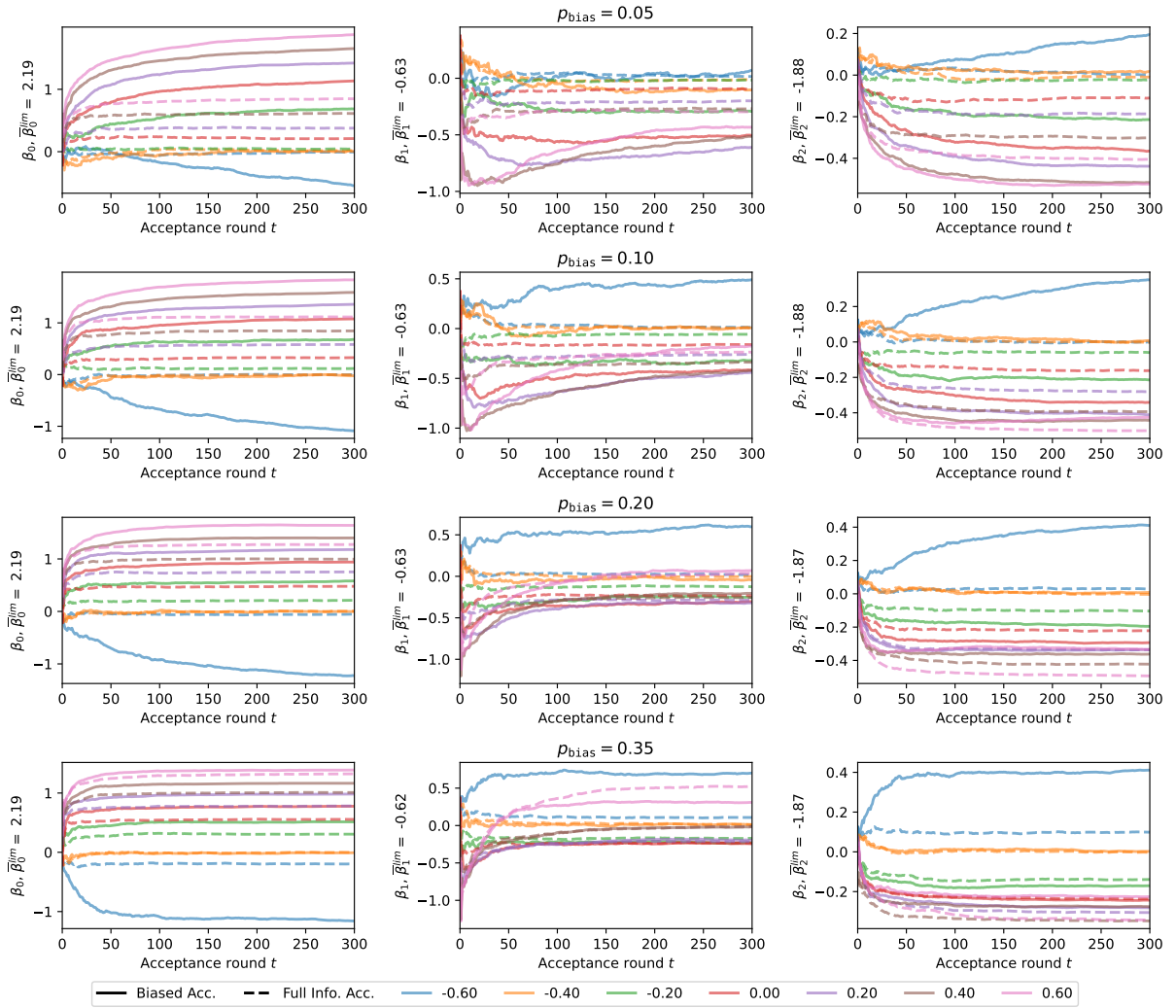


Figure 10: Percentual deviation of Accepts based LR parameters to full information parameters under MNAR based on optimal-ROC point thresholds

The most notable cases occur for $\sigma_{mh} \in \{-0.2, -0.4, -0.6\}$. In the base simulation ($\sigma_{mh} = -0.4$), both $\hat{\beta}_k^f$ and $\hat{\beta}_k^b$ stabilize at values very close to the full-information

parameters, apparently even faster than in the MAR case. A possible explanation is that the hidden factor contains little additional information about Y beyond the model features, making an MNAR mechanism based on X_h closer to one based on X_m alone.

A similar pattern is observed for $\sigma_{mh} \in \{-0.2, -0.6\}$, where the deviations of $\hat{\beta}_k^f$ also stabilize close to zero. Recall from Figure 2 that the corresponding differences between the restricted and full Bayes error rates, $P_e^m - P_e^*$, are relatively small. This relationship is not fully consistent across the two cases, however: $\sigma_{mh} = -0.6$ has a larger Bayes-error difference than $\sigma_{mh} = -0.2$, while the deviations of $\hat{\beta}_k^f$ are smaller. Moreover, $\sigma_{mh} = -0.6$ is the only configuration in which the deviations of $\hat{\beta}_k^b$ show little evidence of stabilization. I do not identify a clear explanation for this behavior from the current simulations, so I treat it as a limitation of the present analysis rather than infer a specific mechanism.

This analysis yields several conclusions for the RI problem. First, even under MAR, convergence of the LR estimates can require substantially more accepted observations than are available in the early rounds of the simulation to generate estimates close to the true population values. The convergence of $\hat{\beta}_k^f$ towards the full-information estimates, however, indicates that a more representative accepted sample can alleviate this problem. In particular, one does not necessarily need labels for the entire applicant population: labels for accepts generated by a more representative acceptance policy may already improve scorecard estimation.

This observation provides motivation for investigating partial debiasing methods such as BASL (Kozodoi et al., 2025) and CI-EX (Ribeiro & Raimundo, 2025). In the present setting, such methods could be useful if they recover sufficiently accurate information in regions of the applicant population that the current policy rarely accepts and a full-information policy would accept. This interpretation remains a hypothesis, since the present simulation does not evaluate those RI methods directly.

5.2.3 Behavior of surprise

Having gained an understanding of how the acceptance bias evolves over time, I can now test whether *surprise* $\eta^{(t)}$ detects this behavior. Recall that surprise is defined as the difference between the historical performance estimate and the realized performance on future accepts. I first consider the full-information policy, which provides the closest setting to the unsurprising benchmark described above.

For this comparison, the cross-validation procedure is adapted so that the historical performance estimate is based on the full-information data while the evaluation population corresponds to applicants accepted by the full-information policy. This provides an oracle benchmark of what would happen if using an RI method one could obtain full-information estimates of the LR parameters and would evaluate them using only the accepts of a policy using such a RI methodology. The resulting estimate is then compared with the realized performance on accepts of the subsequent round.

When calculating surprise, particularly for the accepts-only policy, some rounds contain very few or no accepted bads, making the AUROC and KS statistic unstable or undefined. I therefore calculate realized performance over the next W_f acceptance rounds, which increases the number of observed bad outcomes. Windows containing fewer than four accepted bads are assigned `nan` and excluded from the analysis. This requires the additional

assumption that the acceptance policy remains approximately stable over the window. The preceding analysis suggests that material changes in the acceptance dynamics occur over a longer horizon, but the choice of W_f remains a heuristic.

In Figure 11, the surprise is measured using AUROC when the threshold is selected by the optimal-ROC point and using the KS statistic when the threshold is selected by the KS criterion. The lower x-axis reports the number of generated applicants, while the upper x-axis reports the cumulative number of accepts under the corresponding policy. Because the initial round contains 6000 applicants and each subsequent round adds exactly 1000 applicants, the generated sample size also identifies the acceptance round. For example, 93,000 generated observations correspond to round 87.

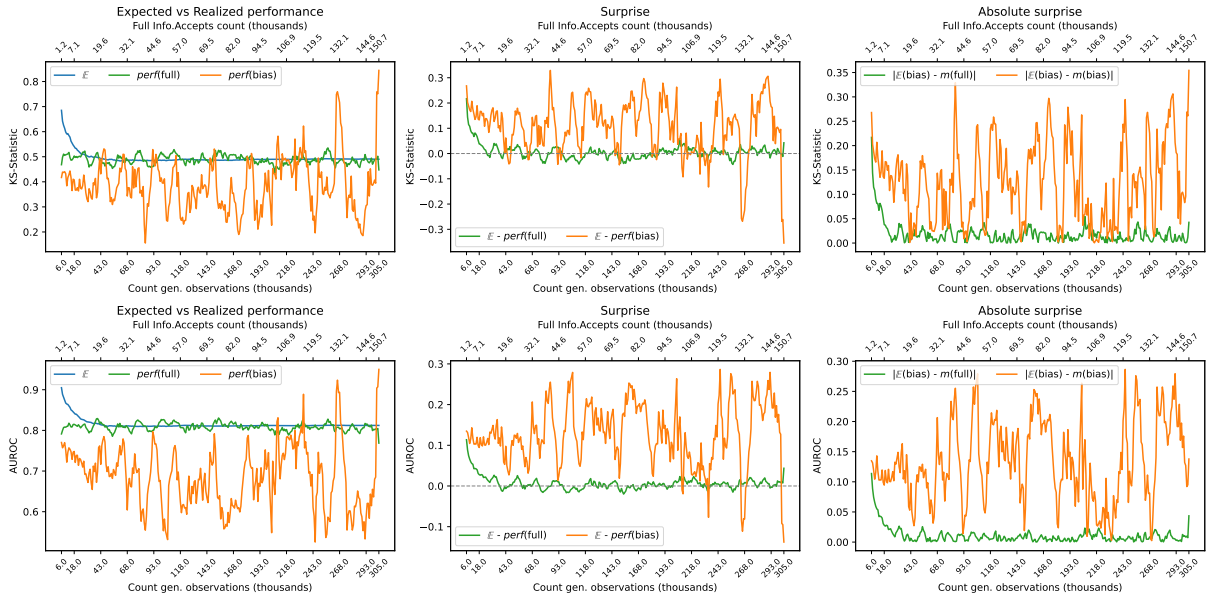


Figure 11: Surprise diagnostics of full information policy with $W_f = 7$

In the left-most panels of Figure 11, the blue line shows the historical performance estimate of the full-information scorecard over its corresponding accepts. The green line shows its realized performance over the subsequent $W_f = 7$ rounds of full-information accepts,⁴¹ while the orange line shows the performance of the same scorecard on accepts from the accepts-only policy. The middle panels report the corresponding differences between the historical estimate and the two realized performance measures, and the right panels show their absolute values.

The results are consistent with the intuition developed in Section 5.1.2. For the full-information policy, surprise (green line in the middle panels) is small after the initial rounds and remains close to zero. In contrast, the performance on accepts from the accepts-only policy differs much more strongly from the historical estimate and is substantially more volatile. This is consistent with the previously observed distributional changes and with the stability of the full-information scorecard and acceptance region. However, the discrepancy does not systematically increase with the acceptance round, as the original

⁴¹This value was selected because it was the smallest value for which the realized performance could be calculated at almost all acceptance rounds.

hypothesis would suggest. At most, the volatility appears to increase in later rounds. The high volatility may be related to the increasingly imbalanced accepted samples, and the results for $W_f = 10$ do not materially change this conclusion.

The same qualitative pattern is obtained when increasing p_{bias} , although the discrepancy between historical and realized performance on accepts-only samples becomes smaller. A plausible explanation is that the overwrite mechanism makes the accepted samples of the two policies more similar, as observed in the preceding analysis. Thus, Figure 11 provides evidence that the full-information policy behaves as an unsurprising benchmark and that the accepts-only policy produces the distributional differences that motivate the surprise diagnostic.

In Figure 12, the same analysis is performed using the scorecard estimated from the accepts-only policy. The surprise (orange line in the middle panel) for this scorecard is more volatile than for the full-information benchmark. During the early and middle rounds, positive surprise is common. Because larger AUROC and KS values indicate better performance, this means that the historical performance estimate tends to be overoptimistic relative to performance on subsequent accepts. This direction is consistent with the overoptimistic evaluation reported in the credit-scoring literature under acceptance-driven selection (Kozodoi et al., 2025).

Later in the simulation, surprise fluctuates around zero. This may reflect the stabilization of the acceptance process in a relatively narrow region of the applicant population. The same pattern becomes somewhat clearer for $W_f = 10$. The performance of the accepts-only scorecard on full-information accepts exhibits smaller discrepancies, although it is difficult to determine from these results whether this reflects a substantive effect or sampling variation.

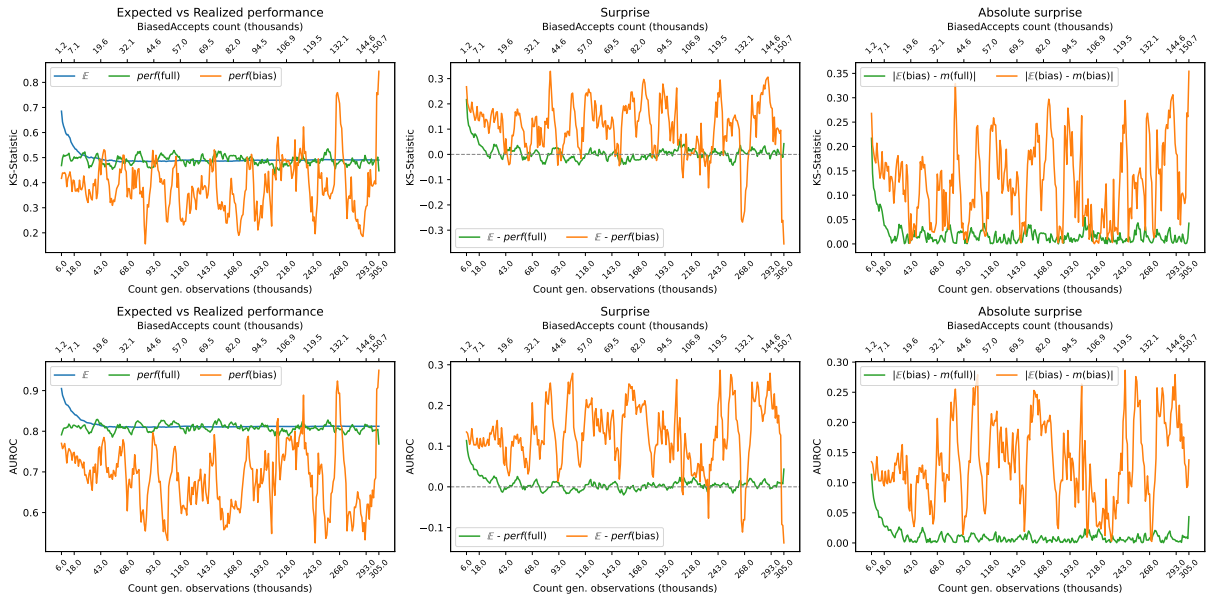


Figure 12: Surprise diagnostics of accepts based policy with $W_f = 7$

Increasing p_{bias} produces two notable effects. First, surprise starts fluctuating around zero earlier, which is consistent with the earlier stabilization of the acceptance rate, bad

rate among accepts, and LR coefficients. Second, for the largest overwrite proportions, $p_{\text{bias}} \in \{0.20, 0.35\}$, the discrepancy between historical and realized performance on full-information accepts becomes larger than the surprise observed for the accepts-only policy. This suggests that the relatively small deviation values of the latter may partly result from the small number of future accepted observations used to estimate realized performance.

The other σ_{mh} configurations show the same qualitative behavior. The initially large surprise of the accepts-only policy followed by stabilization around zero becomes more pronounced as p_{bias} increases, consistent with the stronger stabilization of the LR parameter deviations observed above. For the full-information policy, surprise remains close to zero except in configurations where the restricted and full Bayes error rates differ more strongly. This is consistent with the fact that the full-information policy still estimates the scorecard from X_m only: when X_h contains more additional information about Y , model misspecification can itself contribute to the observed discrepancy.

6 Conclusion

In this thesis, I studied reject inference as a dynamic selective-observation problem. The literature review showed that RI methods generally require additional identifying assumptions or information beyond the observed outcomes of accepted applicants and their covariates. This motivated a more explicit formulation of the RI problem in which the acceptance mechanism is part of the data-generating process rather than a static source of missingness. The measure-theoretic formulation makes the role of the acceptance policy, the observed and hidden variables, and the temporal feedback between successive rounds explicit, avoiding having hidden assumptions as it was often found in the literature review.

Based on this formulation, I developed a simulation framework in which the applicant population remains stationary while the acceptance policy is re-estimated from previously accepted applicants. This isolates the feedback generated by selective observation from exogenous population drift. Within this framework, I formulated and heuristically justified the hypothesis that acceptance bias can become observable through changes in the accepted population and through differences between historically estimated and subsequently realized model performance.

The numerical analysis provides evidence for the hypothesized overfocusing mechanism. In the accepts-only policy, the acceptance threshold declines over successive rounds and the acceptance rate correspondingly contracts, even though the underlying applicant population is unchanged. The accepted population therefore becomes increasingly concentrated in a region associated with lower predicted default risk. This behavior is observed under both MAR and the considered MNAR configurations and is consistent with a self-reinforcing selection mechanism.

The analysis of the LR parameters provides a second perspective on the same effect. The estimates obtained from the accepts of the full-information policy stabilize quickly, whereas the accepts-only estimates can remain substantially farther from the full-information parameters even when their training samples become large. Under MNAR, some configurations additionally produce persistent deviations from the full-information benchmark. These results should not be interpreted as a proof that the LR estimators cease to be consistent: the simulation is finite and does not establish their asymptotic behavior.

They do, however, show that increasing the number of accepted observations alone is not sufficient to eliminate the discrepancy generated by the evolving acceptance process in the simulated setting, at least given the sample sizes studied in this simulation study.

The third part of the analysis investigated *surprise* as a possible diagnostic signal of this bias. The full-information benchmark behaves as expected: its surprise remains comparatively close to zero once the initial rounds have passed. The accepts-only policy, in contrast, produces larger and more volatile surprise, with positive values occurring particularly during the early rounds. This is consistent with an overoptimistic historical performance estimate. However, the absolute surprise does not increase monotonically with the acceptance rounds, and its volatility prevents it from being interpreted as a reliable quantitative measure of acceptance bias in the present setting. The hypothesis is therefore only partially supported: surprise contains information about the changing selection process, but its magnitude is not sufficiently stable to serve as a standalone measure of acceptance bias.

Several limitations qualify these conclusions. First, the simulation uses a deliberately simple Gaussian DGP with a small number of observable and hidden features. Second, the MNAR mechanism is represented by a specific overwrite rule whose increasing contribution can make the accepts-only and full-information policies more similar. Third, the realized surprise depends strongly on the number of accepted bads. This motivated the use of a future window, which reduces sampling variation but also reduces temporal resolution and requires a stability assumption over the window. Excluding windows with very few accepted bads also remove further observations from highly selective periods. Finally, the analysis does not evaluate the performance of a specific RI method. The results therefore provide evidence for a dynamic bias mechanism and for the potential observability of its consequences, rather than an identified RI procedure.

Future work can build on this framework in four directions. First, the behavior of surprise could be studied more formally under a stable MAR process to characterize its finite-sample variation and to distinguish genuine selection-induced changes from ordinary estimation noise. Theoretical work could benefit from the proposed measure theoretic characterization. For further simulation based research probably other metrics besides AUROC and the KS statistic should be included that may be less sensitive to highly imbalanced data. Second, the proposed framework can also work to study evaluation methods developed for RI such as the Bayesian Evaluation framework proposed by Kozodoi et al. (2025), as other diagnostics signals of the acceptance induced bias. Third, the dynamic formulation could be extended to more realistic MNAR and manual-overwrite mechanisms and eventually connected to RI procedures that use the observed dynamics explicitly.

The contribution of this thesis is threefold. First, it formulates RI as a dynamic selective-observation process in which current decisions determine the labeled data available for future decisions.

The main contribution of this thesis is therefore not a new RI algorithm, but a new perspective of reject inference that makes the temporal feedback of selective observation explicit and empirically testable. The numerical results indicate that this feedback can progressively alter the accepted population and the scorecards estimated from it, even when the underlying applicant population is stationary. One important contribution thereby is the presented framework on how to observe this bias along different dimensions

in a simulation environment. Surprise captures part of this effect, but the present analysis also shows that observability is not equivalent to reliable measurement: a signal can contain information about acceptance bias proxied as distributional changes in the accepts population, without providing a stable estimate of its magnitude. This distinction provides the main point of departure for future work on RI methods that use the dynamics of the acceptance process as an additional source of information.

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7 Appendix

7.1 Acknowledgement of AI usage

Generative Artificial Intelligence were used as interactive assistants for discussing statistical concepts, reviewing mathematical arguments, suggesting improvements to exposition, identifying relevant literature, reviewing software implementations and help in generating code. The used AI's were

- ChatGPT (OpenAI, 2026a), mostly GPT 5.1
- Claude (Anthropic PBC, 2026), mostly Sonnet 4.6
- Microsoft Copilot (Microsoft, 2026), Microsoft Prometheus model powered by ChatGPT
- Perplexit (Perplexity AI, Inc., 2026)
- GitHub Copilot (GitHub & OpenAI, 2026)
- DeepAI (DeepAI, 2026)
- NotebookLM (Google, 2024)

A close bookkeeping of the usage can be seen on [GitHub](#)

7.2 Guide on mathematical notation

Some of the notation used can be misunderstood. Therefore some help to understand certain notations is provided in Table 2. Thereby let $M \in \mathbb{R}^{n \times m}$ with $n, m \in \mathbb{N} \setminus \{1\}$, S some set, A a tensor of rank greater 0 and a any “scalar” element (a rank 0 tensor).

Table 2: Some notations

Notation	Meaning
$ M $	Given $n = m$ the determinant of M
$ S $	Cardinality of S (amount of elements)
$(s \in S \text{condition})$	Given S containing tensors, a way to define a tensor of one higher rank, new dimension appended at the beginning.
$A[\dots]$	Analogous on how one defines a set Slice of A following notational numpy's conventions , see the example for slicing below
$M^\top$	Transpose of M
$\text{nrows}(M)$	Dimensionality of first dimension. In this case n
$\vec{a}_n$	A vector with n entries, all of them equal to a

Some examples to understand all the slicing operations done in the thesis are now provided:

- Let $M' = M[1 :, : m - 1]$ then $M' \in \mathbb{R}^{n-1 \times m-1}$ where M' contains the last $n - 1$ rows of M and its first $m - 1$ columns.
- $M'[0, :] = M'[0] \in \mathbb{R}^{m-1}$ is the vector (rank 1 tensor) of the first row of M'
- $M'[:, 0] \in \mathbb{R}^n$ is the vector (rank 1 tensor) of the first column of M'
- Let $0 < k < n - 1$ and $\tilde{M} = (M[:, j] | j \in \mathbb{N}, j < k)$, then $\tilde{M} \in \mathbb{R}^{k \times n-1}$
- $M[0, 0] \in \mathbb{R}$ is the scalar at the first row and column of M

7.3 Mathematical derivations/proofs

7.3.1 Set of positive definite variance covariance matrices for the simulation

As stated in Section 4.1, the common covariance matrix Σ has the general shape (for the 3 variables case)

$$\Sigma = \begin{pmatrix} 1 & \sigma_{m,1,2} & \sigma_{mh} \\ \sigma_{m,1,2} & 1 & \sigma_{mh} \\ \sigma_{mh} & \sigma_{mh} & 1 \end{pmatrix} \quad (25)$$

For notational simplicity let here $a \equiv \sigma_{m,1,2}$ and $b \equiv \sigma_{mh}$.⁴² The space of positive definite covariance matrices following Eq. 25 can be found using Sylvester's criterion⁴³ and is fully determined by the admissible tuples $(a, b) \in \mathbb{R}^2$.

Using the simplified notation, the covariance matrix can be written as

$$\Sigma = \begin{pmatrix} 1 & a & b \\ a & 1 & b \\ b & b & 1 \end{pmatrix}. \quad (26)$$

The first leading principal minor always has a positive determinant since $\det(1) = 1 > 0$. The second leading principal minor is

$$\det \begin{pmatrix} 1 & a \\ a & 1 \end{pmatrix} = 1 - a^2. \quad (27)$$

Therefore, the second condition requires

$$1 - a^2 > 0, \quad (28)$$

which is equivalent to

$$-1 < a < 1. \quad (29)$$

Finally, the determinant of the full matrix in Eq. 26 is

$$\det(\Sigma) = \begin{vmatrix} 1 & a & b \\ a & 1 & b \\ b & b & 1 \end{vmatrix}. \quad (30)$$

Subtracting the second row from the first row leaves the determinant unchanged and yields

⁴²Note: from here on the derivation has been fully AI-generated with minor changes by me. The facts were checked upon by me. See OpenAI (2026b) for the reference to the chat.

⁴³I.e. if all leading principal minors have a positive determinant, then the matrix is positive definite

$$\det(\Sigma) = \begin{vmatrix} 1-a & a-1 & 0 \\ a & 1 & b \\ b & b & 1 \end{vmatrix}. \quad (31)$$

Factoring out $(1-a)$ from the first row gives

$$\det(\Sigma) = (1-a) \begin{vmatrix} 1 & -1 & 0 \\ a & 1 & b \\ b & b & 1 \end{vmatrix}. \quad (32)$$

The cofactor expansion results in

$$\det(\Sigma) = (1-a) \left[\begin{vmatrix} 1 & b \\ b & 1 \end{vmatrix} + \begin{vmatrix} a & b \\ b & 1 \end{vmatrix} \right]. \quad (33)$$

Evaluating the two 2×2 determinants yields

$$\det(\Sigma) = (1-a) [(1-b^2) + (a-b^2)]. \quad (34)$$

After simplification,

$$\det(\Sigma) = (1-a)(1+a-2b^2). \quad (35)$$

By Eq. 29, the factor $(1-a)$ is strictly positive. Consequently, the third condition of Sylvester's criterion is

$$1+a-2b^2 > 0. \quad (36)$$

Solving for b gives

$$b^2 < \frac{1+a}{2}, \quad (37)$$

Combining Eq. 29 and Eq. 37, the set of positive definite covariance matrices of the form Eq. 25 is characterized by

$$\left\{ (a, b) \in \mathbb{R}^2 : -1 < a < 1, b^2 < \frac{1+a}{2} \right\}. \quad (38)$$

7.3.2 On the Bayes Classifier with idiosyncratic shocks

7.3.2.1 Decision boundary

The decision boundary between two classes is defined as the equal probability region for both. For our case with two classes there is only one boundary and it is given by

$$\begin{aligned}\mathbb{P}(Y = b|X = x) &= \mathbb{P}(Y = g|X = x) \\ \Leftrightarrow \mathbb{P}(Y = b|X = x) &= \frac{1}{2}\end{aligned}$$

Plugging in Eq. 17 we get

$$\begin{aligned}\pi_\epsilon p_{\tilde{Y}|X}(b, x) + (1 - \pi_\epsilon)\pi_{\epsilon,b} &= \frac{1}{2} \\ \Leftrightarrow p_{\tilde{Y}|X}(b, x) &= \underbrace{\frac{\frac{1}{2} - (1 - \pi_\epsilon)\pi_{\epsilon,b}}{\pi_\epsilon}}_{=:\xi}\end{aligned}\tag{39}$$

Note that Eq. 39 shows how considering the idiosyncratic shocks only moves the decision boundary. If there were no shocks ($\pi_\epsilon = 1$) or their effect was symmetric ($\pi_{\epsilon,b} = \frac{1}{2}$) then $\xi = \frac{1}{2}$, which equals the decision boundary based on a classifier ignoring the shocks.

Using Eq. 16, Eq. 15 and Eq. 14 we get

$$p_{\tilde{Y}|X}(b, x) = \frac{\phi_{X_b}(x)\pi_b}{\pi_b\phi_{X_b}(x) + \pi_g\phi_{X_g}(x)} = \xi$$

and using $\pi_g = 1 - \pi_b$

$$\Leftrightarrow \pi_b(1 - \xi)\phi_{X_b}(x) = \xi(1 - \pi_b)\phi_{X_g}(x)$$

which allows for a odds ratio perspective

$$\Leftrightarrow \frac{\phi_{X_b}(x)}{\phi_{X_g}(x)} = \frac{\xi(1 - \pi_b)}{\pi_b(1 - \xi)} =: c\tag{40}$$

Since we assume equal covariances matrices we can get from taking logs the familiar result (see e.g. Hastie et al., 2009, p. 108)

$$x^\top \Sigma^{-1}(\mu_b - \mu_g) - \underbrace{\frac{1}{2}(\mu_b + \mu_g)}_{=:\bar{\mu}} \Sigma^{-1}(\mu_b - \mu_g) = \log c$$

After factorizing and using the symmetry of Σ^{-1} we get

$$(\mu_b - \mu_g)^\top \Sigma^{-1}(x - \bar{\mu}) = \log c\tag{41}$$

7.3.2.2 Bayes error rate

A perfect Bayes classifier considering the idiosyncratic shock as defined in Eq. 17 has an analytical solution for the Bayes error. To get to this expression, I use the results of Ripley (1996).

For any $\omega \in \Omega$, the decision to classify an applicant as bad with characteristics $x := X(\omega)$ occurs if

$$\mathbb{P}(Y = b|X = x) > \mathbb{P}(Y = g|X = x)$$

which following the same steps as before we get an expression like Eq. 41 that we can use to also define a random variable

$$A(\omega) := (\mu_b - \mu_g)^\top \Sigma^{-1}(x - \bar{\mu}) > \log c$$

where, following Ripley (1996),

$$A|Y = b \sim \mathcal{N}\left(\frac{1}{2}\delta^2, \delta^2\right), A|Y = g \sim \mathcal{N}\left(-\frac{1}{2}\delta^2, \delta^2\right)$$

with

$$\delta = \sqrt{(\mu_b - \mu_g)^\top \Sigma^{-1}(\mu_b - \mu_g)}$$

Therefore the decision function d_B of the bayes classifier can be written as

$$d_B(x) = \begin{cases} b & A > \log c \\ g & A \leq \log c \end{cases}$$

The bayes error rate E_B is then given by

$$\begin{aligned} E_B &= \mathbb{P}(Y = b)\mathbb{P}(A \leq \log c|Y = b) + \mathbb{P}(Y = g)\mathbb{P}(A > \log c|Y = g) \\ &= \pi_b \Phi\left(-\frac{1}{2}\delta + \frac{1}{\delta} \log c\right) + (1 - \pi_b) \Phi\left(-\frac{1}{2}\delta - \frac{1}{\delta} \log c\right) \end{aligned} \quad (42)$$

7.3.3 A possible route towards a formal justification of surprise as a valid signal

The argument in Section Section 5.1.2 is intentionally heuristic. Nevertheless, it is useful to outline how one could attempt to formalize the claim that surprise should vanish in a stable MAR environment, which is the sole purpose of this appendix.

Assume the conditions from Section Section 5.1.2 hold. Define the acceptance region at round t as

$$A_t := \{x \in \mathbb{R}^{f_m} \mid \hat{s}_t(x) > c_t\}.$$

The corresponding distribution of accepted applicants is then

$$\mathbb{P}_{Y,X|X \in A_t}(E) = \frac{\mathbb{P}((Y, X) \in E, X \in A_t)}{\mathbb{P}(X \in A_t)}$$

for measurable $E \in 2^{E_Y} \otimes \mathcal{B}(\mathbb{R}^f)$.

Suppose additionally that the estimation method is unbiased in the sense of this thesis, namely that given enough representative data, the estimated scorecard $\hat{s}_t$ provides a good approximation of a scoring proportional to $\mathbb{P}(Y = b \mid X)$. Under suitable regularity conditions and growing sample sizes, one could then expect the sequence of scorecards $\hat{s}_t$ to stabilize around some limiting scorecard $s^* \propto \mathbb{P}(Y = b \mid X)$.

Likewise, if the cutoff selection criterion remains unchanged across rounds, one could expect the sequence of cutoffs c_t to stabilize around a limiting value c . This motivates the introduction of the limiting acceptance region

$$A := \{x \in \mathbb{R}^{f_m} \mid s^*(x) > c\}.$$

Heuristically, stabilization of $\hat{s}_t$ and c_t suggests that the acceptance regions A_t become increasingly similar to A . As a consequence, one would expect the distributions of accepted applicants

$$\mathbb{P}_{Y,X|X \in A_t}$$

to become increasingly similar across rounds as well.

Although turning this intuition into a rigorous theorem would require additional assumptions on the estimation method, the applicants distribution and the acceptance regions, the central idea is simple: in a stationary MAR environment the acceptance mechanism itself should eventually stabilize. Once this occurs, both the training data and future accepted applicants are generated from nearly the same distribution.

Under such circumstances, one would expect the performance estimated through cross-validation on historical accepts to be close to the performance observed on future accepted applicants, implying only small values of the surprise signal $\eta^{(t)}$.

7.4 Acceptance rate and bads rate among accepts for other σ_{mh}

The following graphics are based on Figure 7 with different σ_{mh} configurations.

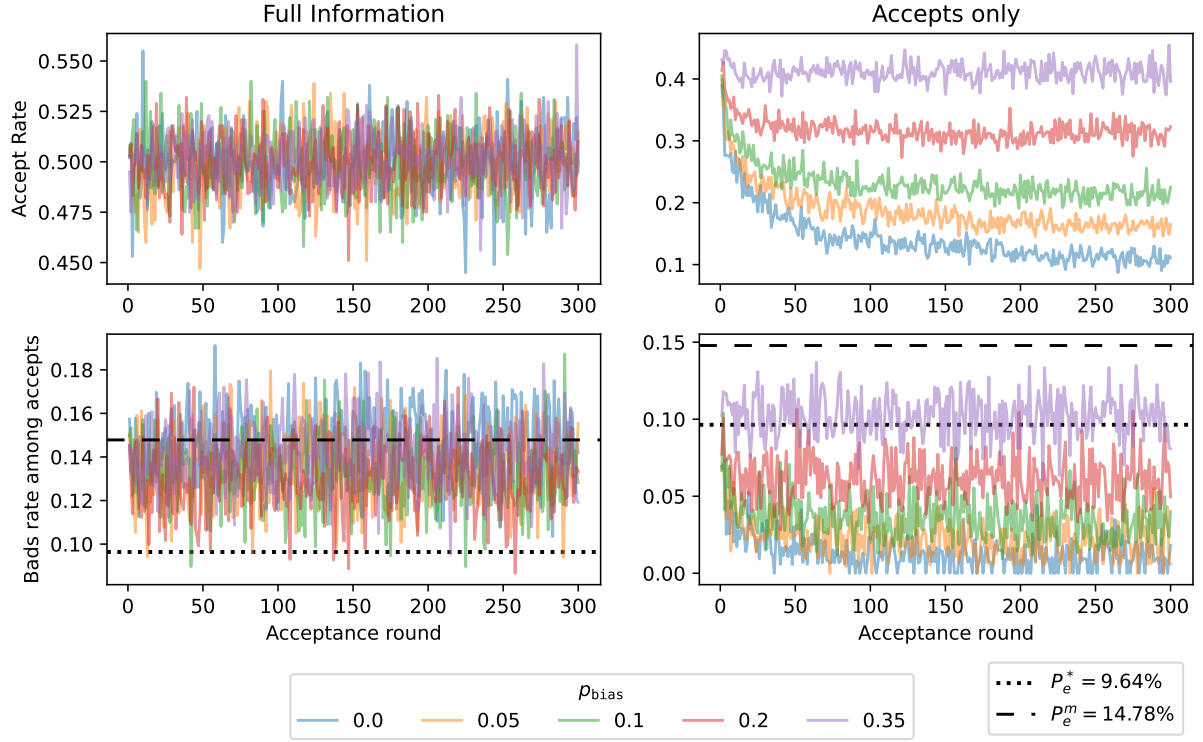


Figure 13: Development of acceptance rates and bad rates among accepts across p_{bias} with $\sigma_{mh} = 0.2$

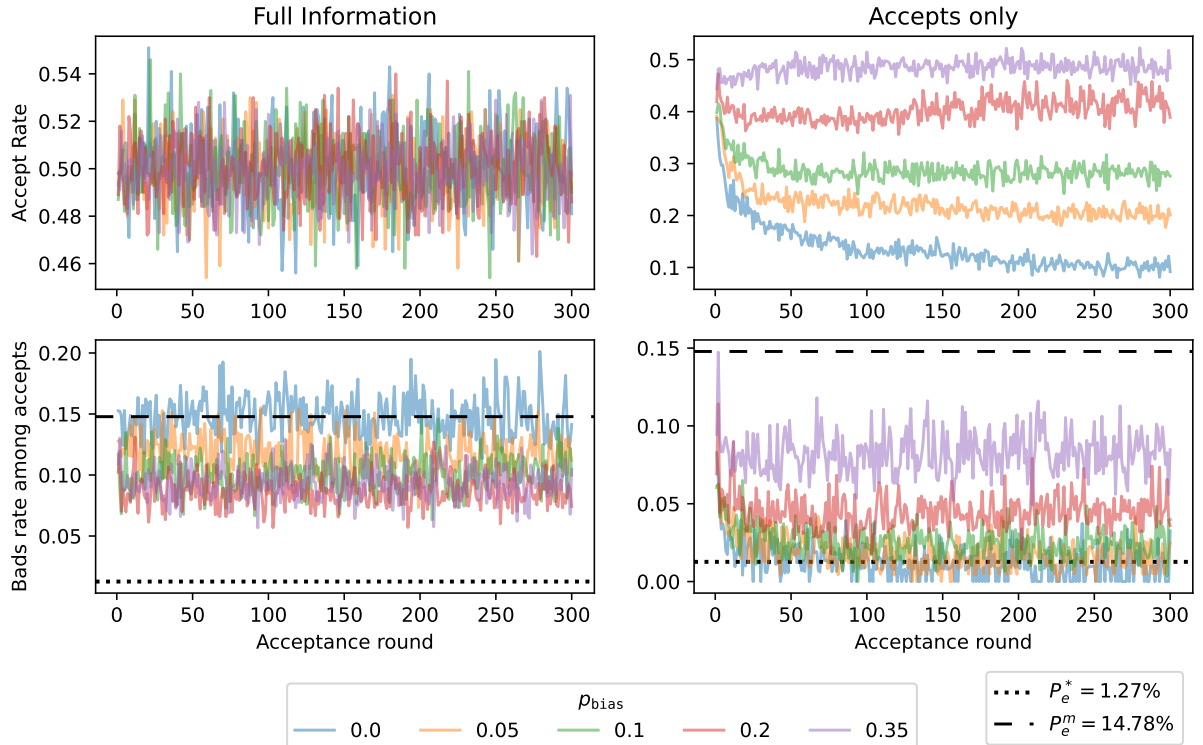


Figure 14: Development of acceptance rates and bad rates among accepts across p_{bias} with $\sigma_{mh} = 0.6$